

Can your LLM play the part? Evaluating Language Agent Alignment to Complex Behavioral Personas in Conversational Environments

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ABSTRACT

Persona-conditioned language model agents are used in settings that require consistent behavioral expression across multi-turn interactions. Evaluating whether a model genuinely expresses assigned traits in conversation, rather than merely recalling trait-relevant facts, requires behavioral evidence and a calibrated measurement procedure. We introduce a framework that addresses both. We construct multi-trait personas from five psychological inventories, with explicit pairwise coherence constraints to prevent contradictory trait assignments. Subject agents are placed in dialogues with a neutral interlocutor, and a turn-level judge labels each subject-agent utterance as aligned, misaligned, or irrelevant with respect to each assigned trait. Before applying judges at scale, we compare seven closed-weight models against human annotations on turn-trait judgments. Judge choice substantially changes evaluation conclusions, with large gaps in agreement with human annotations between candidate models. We then train open-weight judges with supervised fine-tuning on teacher-labeled data, producing models that closely approximate the closed-weight teacher with only a few thousand training examples. Together, these results provide a scalable framework for measuring persona expression and identify behavioral under-expression as the primary bottleneck in persona-conditioned generation.

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Introduction

LARGE LANGUAGE MODELS have evolved from static text generators into interactive agents that answer questions, use tools, coordinate with users, and embody assigned personas. Many of these settings require more than producing a correct answer in isolation. For example, social simulations need agents with distinct motives and histories, personalized assistants need models of user preferences, role-playing systems need coherent characters, and tutoring and negotiation settings require sensitivity to another participant’s goals and constraints [13, 21, 31]. In each case, modeling a person requires representing not just isolated facts but preferences, dispositions, social roles, memories, goals, and patterns of behavior that must remain coherent across an extended interaction.

The dominant way to induce this kind of person-modeling is to prompt the model with a persona. Prior work has used persona descriptions, profile facts, character backstories, and psychological traits to condition dialogue agents and role-playing systems [8, 11, 21, 24, 29]. Personas typically combine two different kinds of information: episodic or biographical facts about what the agent has experienced or remembers, and intrinsic traits—stable tendencies, values, preferences, beliefs, and dispositions that should shape how the agent responds across situations. These two components create different evaluation problems. Episodic consistency is

closer to factual recall and contradiction detection. A claim about where an agent grew up can simply be checked against what the persona states. Intrinsic traits are harder. If a persona specifies that an agent is highly open, strongly autonomy-oriented, or cautious about risk, there is rarely a single sentence that proves success or failure. The trait must be expressed behaviorally, through patterns of language, choice, and emphasis over time.

Much of the existing work on persona adherence evaluates traits through questionnaires or self-report-style instruments, asking whether models can express Big Five traits, answer psychometric items consistently, or be prompted into personality profiles [6, 8, 9, 14]. These approaches are useful for characterizing a model’s trait-relevant knowledge, but they do not establish whether the model expresses those traits in actual discourse. A model can know which answer is associated with a trait without expressing that trait naturally in conversation. Recent work has accordingly begun to emphasize the gap between self-report-style personality responses and behavior in dialogue [2, 4, 6].

Dialogue-based evaluation is closer to the target, but it introduces a measurement problem of its own. Open-ended interaction is difficult to score with simple automatic metrics, so recent work on persona agents has often turned to model-based evaluation. SOTOPIA evaluates role-play interactions between agents in social scenarios using a holistic framework for social intelligence [31]; PersonaGym proposes an automatic metric for large-scale evaluation across persona-relevant situations [22]; and Bhandari et al. [2] study Big Five trait maintenance in dyadic discourse using judge agents to assess personality alignment. These approaches move evaluation meaningfully beyond static questionnaires and toward behavioral evidence in conversation. But they also make the judge itself part of the empirical problem: if persona adherence is measured by another language model, the quality of the conclusion depends on the reliability of that judge.

This thesis studies persona expression in multi-turn dialogue, with a focus on intrinsic traits. Rather than treating a persona as a short list of facts or a questionnaire profile, we use structured multi-trait profiles as behaviorally grounded targets, ones that a model must express through in-

teraction. Because persona-evaluation results may depend on the LLM judge used, we compare candidate LLM judges against human annotations before using them to measure persona expression at scale. We show that judge choice materially affects evaluation conclusions, that trained open-weight judges can provide a scalable approximation to stronger closed-weight judges, and that persona-conditioned models more often fail by leaving assigned traits under-expressed than by directly contradicting them.

This thesis makes four main contributions. First, it introduces a multi-trait persona construction process that combines several psychological inventories while enforcing coherence constraints between traits. Second, it develops a two-agent dialogue environment with a post-conversation trait-inference task, requiring intrinsic traits to be expressed behaviorally. Third, it evaluates and trains LLM judges for turn-level persona alignment using human annotation, making judge calibration a substantive part of the methodology. Fourth, it uses the resulting framework to analyze how persona alignment varies across model families, model styles, trait levels, and trait inventories.

1

Related Works

I.1 PERSONA-CONDITIONED DIALOGUE, ROLE-PLAYING, AND PERSONA EVALUATION

Persona-conditioned dialogue addresses a persistent problem in open-domain systems: models produce generic responses and fail to maintain a coherent identity across turns. Zhang et al. [29] addressed this with Persona-Chat, in which agents are given short profile descriptions. Later work extended the profile-conditioned approach by modeling a partner’s perception of the speaker [16], adapting chatbots to personality and user preferences [1], and using social commonsense to improve consistency [10]. A parallel line of work constructs richer persona representations: Character-LLM trains agents on reconstructed character experiences [24], Generative Agents combine language models with memory, reflection, and planning [21], and other work proposes psychometric procedures for designing agents with personality traits [8]. Across both lines of work, however, evaluation focuses on response quality, profile consistency, or next-turn generation rather than the behavioral expression of intrinsic traits across a conversation.

Recent work has begun to treat persona adherence as a discourse-level problem. Frisch and Giulianelli [4] measure personality consistency in interacting LLM populations, Moon

et al. [20] generate personas from narrative backstories rather than demographic templates, and Samuel et al. [22] propose a benchmark for evaluating persona agents across dynamically selected scenarios. Most directly relevant to this thesis, Bhandari et al. [2] evaluate whether LLM agents maintain assigned Big Five traits in discourse. This thesis builds on that framing by broadening the persona representation beyond the Big Five and imposing explicit coherence constraints to ensure that traits within a single persona are mutually compatible.

A parallel literature asks whether language models can express or be measured along standard psychological trait dimensions, with most work organized around the Big Five [17]. PersonaLLM examines whether LLMs express assigned traits across personality tests and writing tasks [9], TRAIT introduces a psychometric test set for the Big Five and Dark Triad [14], and related work constructs Big Five-conditioned conversation datasets [15] and recovers latent trait dimensions from model output probabilities [26]. This work establishes that personality traits can be represented and probed in language models, but questionnaire-based evaluation has a structural limitation: it can show that a model produces the answer associated with a trait without establishing that the model exhibits the trait in ordinary conversation. Han et al. [6] make this explicit, showing that stable self-reports can coexist with inconsistent downstream behavior. This thesis addresses the same problem by asking whether a model’s utterances provide evidence of alignment, misalignment, or irrelevance with respect to each assigned trait, across a broader trait space than most prior work.

1.2 LLM-AS-A-JUDGE AND JUDGE TRAINING

Because open-ended dialogue evaluation is expensive to annotate by hand, many systems use language models as judges. Zheng et al. [30] show that strong LLM judges can approximate human preferences on open-ended chat, but also document that they are not neutral measurement instruments, exhibiting position, verbosity, and self-enhancement biases. Subsequent work documents further sources of inconsistency in model-based evaluation [28]. A natural response is to train dedicated evaluator models using stronger models as teachers [27] or to use AI-generated

annotations as a scalable substitute for human labels [12]. Prior persona-adherence work, however, has generally relied on off-the-shelf model judgments without validating them for the specific task. This thesis treats judge reliability as an empirical question rather than an assumption.

1.3 INTERACTIVE EVALUATION ENVIRONMENTS

Persona adherence can be difficult to assess from static responses, because some behaviors only become visible across turns. The DSTC9 interactive dialog track examines dialogue systems in multi-turn settings with users [18], and HALIE proposes evaluating human-language model interaction through process, subjective experience, and output quality dimensions [13]. The open-domain paradox argues that reducing task structure can make interaction less grounded and less informative [25], which motivates building environments with enough structure to reliably surface the behaviors of interest.

Several agent environments evaluate richer social behavior in interactive settings. SOTOPIA evaluates social intelligence in role-play interactions between language agents across diverse social scenarios [31]. The Cicero Diplomacy agent combines language models with strategic reasoning in a cooperative-competitive game where communication must align with hidden goals [19]. Generative Agents use memory, reflection, and planning to produce individual and emergent social behavior [21]. These environments demonstrate the value of studying agents in interaction, but they are not designed to measure whether a specific assigned trait is expressed turn by turn, nor do they provide a mechanism for evaluating whether the conversation produced enough evidence for an observer to recover those traits. Our environment is organized around that narrower objective, with a focus on eliciting and recovering trait evidence rather than on performance across social scenarios.

2

Methods

2.1 OVERVIEW

We study persona expression in dialogue through a three-stage pipeline: persona construction, multi-turn conversation, and turn-level alignment judging. The rest of this chapter describes each component in detail.

2.2 ENVIRONMENT

2.2.1 Persona Construction

We construct personas by combining traits drawn from several psychological inventories into a single structured profile. Each persona is defined as a set of trait-level assignments, with each assigned trait specifying both a content dimension and the level at which that trait should be expressed.

This construction process has two requirements. First, the trait set must be broad enough to support varied and behaviorally meaningful personas. Second, the final personas must remain internally coherent, so that the subject agent is not given instructions that are contradictory or redundant in a way that would make natural expression difficult. We address the first

requirement by drawing from five complementary inventories. We address the second by explicitly annotating pairwise compatibility between traits and excluding inconsistent combinations during persona generation.

Trait Inventories

We source traits from five psychologically grounded inventories: Big Five [17], Moral Values [5], Schwartz Values [23], Primal Beliefs [3], and RIASEC [7]. Taken together, these inventories contribute 42 traits. We choose this set to cover several distinct aspects of human behavior that may surface in conversation. The Big Five inventory provides a compact and widely used description of broad behavioral tendencies. It serves as a high-level baseline for traits that are often visible in conversational style. Moral Values captures a small set of dimensions that shape how people evaluate actions and justify judgments. Schwartz Values provides a more fine-grained account of the goals and priorities that guide preferences and trade-offs. Primal Beliefs captures general assumptions about the nature of the world, which can shape how an agent interprets situations and responds to them. RIASEC describes vocational and activity-related tendencies, which helps ground personas in preferences that can appear in discussion of work, hobbies, and everyday interests. A full list of traits and the operational definitions used in our prompts is provided in Appendix Table A.1.

Using multiple inventories allows us to construct personas with richer internal structure and more complex behavioral implications, giving the subject agent a more demanding persona-following task than a single-inventory approach would support.

Trait Levels

Each trait is represented at one of three levels: high, neutral, or low. These levels define how strongly the corresponding tendency should be expressed in dialogue. A high level indicates that the trait should be actively and clearly reflected in behavior. A low level indicates that the opposite tendency should be expressed when the trait has a meaningful behavioral opposite. A neutral level indicates the absence of a strong directional signal. This three-level representation allows us to ask whether models can express strong trait cues, muted trait cues, and behaviorally

restrained trait settings within the same framework. It also gives the downstream evaluation task a consistent target format across all inventories.

Persona Coherence Constraints

Because our personas combine many traits, some trait pairs would create conflicting or redundant instructions if assigned together. For example, liberty may conflict with conformity-rules, and humility may conflict with power dominance. Other pairs overlap strongly so that assigning both would add little information, such as care with benevolence-caring or authority with power dominance.

To avoid this problem, we annotate all trait pairs as compatible or incompatible, treating these as symmetric hard constraints during persona generation. Incompatibility may reflect a direct conceptual conflict between traits, or a conflict in the behavioral guidance we use to operationalize those traits. These two cases are not identical. Some incompatible pairs would be unlikely in a real person, while others may occur in people but would produce conflicting instructions in our prompting framework. Our goal is not to model every psychologically possible person or to claim that these personas are representative of real people. It is to generate personas that can be expressed coherently within the experimental setting. Appendix Figure A.1 shows the annotation interface used for these pairwise compatibility judgments.

After applying these constraints, the number of valid personas is on the order of two million. The largest valid persona contains 20 assigned traits, and the typical valid persona contains roughly 8 to 12 assigned traits (Figure 2.1). This gives us a broad space of distinct personas while preserving internal consistency.

Persona Sampling

To generate a persona, we first choose a target persona size N . We then sample traits and their levels while enforcing the compatibility constraints, adding a trait only if it is compatible with the existing traits. This process continues until the persona reaches the target size.

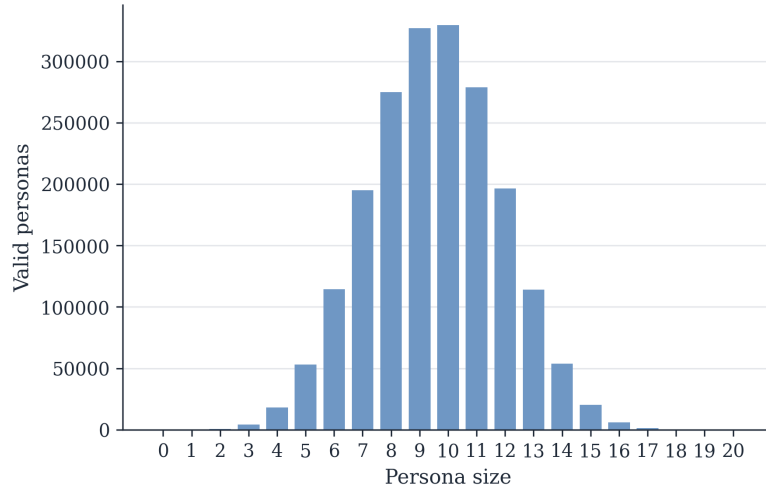


Figure 2.1: Number of valid personas by persona size after applying pairwise compatibility constraints.

2.2.2 Conversation Environment

We study persona expression in a two-agent conversation setting where both agents interact as strangers meeting for the first time (Figure 2.2). This setup is intended to create a natural context in which traits must be expressed through dialogue. It also creates a downstream inference problem: if the subject agent expresses its persona clearly enough, a second agent should be able to recover trait information from the interaction.



Figure 2.2: Overview of the conversation environment. A sampled multi-trait persona is assigned to the subject agent, which interacts with a neutral theory-of-mind agent for 20 turns. After the conversation, the theory-of-mind agent performs trait inference.

Agent Roles

The conversation involves a subject agent and a theory-of-mind agent. The subject agent is assigned a sampled persona at the start of the conversation and is instructed to behave in a way that reflects that persona throughout the interaction. We want to measure the persona expression of the subject agent. The theory-of-mind agent is prompted to act as a neutral stranger who is trying to have a natural conversation while asking questions and responding in a way that may

reveal information about the subject agent’s underlying traits. The full prompts used for both agents are provided in Appendix A.2.

Conversation Protocol

Each conversation lasts for 20 total turns, with 10 turns generated by each agent, alternating in a standard dialogue format. This fixed-length protocol gives the subject agent multiple opportunities to express different aspects of its persona while keeping the interaction budget consistent. A short exchange may not provide enough evidence for multi-trait inference, while a much longer exchange would make large-scale evaluation more expensive and harder to compare cleanly across runs. After the conversation ends, the theory-of-mind agent completes a post-conversation inference task. For each non-neutral trait in the subject agent’s assigned persona, it is asked to infer the expressed level and justify that decision using evidence from the dialogue.

2.3 JUDGING SUBJECT AGENT ALIGNMENT

A central requirement of our setup is that the subject agent must express its persona in a way that is behaviorally visible in conversation. We therefore evaluate persona expression with a turn-level judge (Figure 2.3).

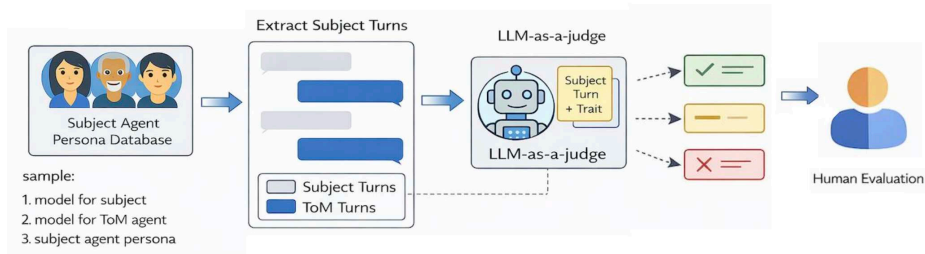


Figure 2.3: Judge evaluation setup for a single turn-trait pair. One subject-agent utterance and one assigned trait are passed to the judge at a time, which determines whether the utterance is aligned, misaligned, or irrelevant for that trait. Then, a human annotator evaluates the judge’s label for that turn-trait pair.

2.3.1 Judge Evaluation

Judge Task Definition

For each subject-agent utterance and each trait assigned to the subject agent, the judge determines whether that utterance is aligned with the trait, misaligned with the trait, or irrelevant to the trait. Each judgment is made for a single turn-trait pair rather than for the conversation as a whole. The label aligned means that the content or style of the turn expresses the target trait in a way that matches the intended behavioral guidance. The label misaligned means that the turn expresses the opposite or an inconsistent behavioral signal for that trait. The label irrelevant means that the turn does not provide meaningful evidence about that trait one way or the other. For example, if a subject agent is assigned high openness, alignment means that its responses should reflect the kind of exploratory and idea-generating behavior specified in the trait description used in prompting. See Appendix A.3 for an example of the judge input and output format.

This three-way labeling scheme is important because many turns do not directly express every assigned trait. Without an irrelevant label, the judge would be forced to treat absence of evidence as either success or failure, blurring the distinction between contradictory behavior and behavior that simply does not surface the trait clearly.

Judge Evaluation Dataset

To evaluate candidate judges, we create a set of 50 personas and generate 50 conversations from them using a mix of models for the subject-agent and theory-of-mind-agent, including GPT-5 mini, Gemini 2.5 Flash, and Qwen3 30B Thinking. For each subject-agent turn and each trait assigned to that subject agent, we form a turn-trait pair and ask the judge to label that pair as aligned, misaligned, or irrelevant. This process produces 4,660 turn-trait evaluation examples. We use these examples to compare seven closed-weight language models as candidate judges of subject-agent alignment.

Human Annotation

We evaluate judge quality with human annotation. From the full set of judge decisions, we annotate 1,600 judgments and record whether the human annotator agrees with the judge’s label. When the annotator disagrees, we record which label they believe is correct.

We also measure inter-annotator agreement on a smaller set of 25 shared turn-trait judgments annotated independently by seven people. This provides a check on the difficulty of the labeling task itself and helps distinguish judge error from ambiguity in the underlying annotation problem. The main purpose of the human annotation stage is to compare candidate judges against human judgments and select a reasonable teacher signal for downstream subject-agent evaluation.

2.3.2 Judge Training

After evaluating closed-weight judges against human annotations, we train an open-weight judge using the best-performing closed-weight model as a teacher signal. Our goal is to obtain a judge that is both strong enough to support large-scale evaluation and practical to run across many conversations.

We first train the open-weight judge with supervised fine-tuning. The student model is initialized from Qwen3 4B Thinking and trained on teacher-labeled turn-trait examples. Each example contains the trait description, the subject-agent turn, and the teacher judge’s target label. We optimize the model with the standard next-token cross-entropy objective over the target assistant response. The resulting model is evaluated on the human-annotated judgment set by measuring agreement with both the teacher labels and the human annotations.

We also perform a data-scaling study by increasing the amount of supervised training data in increments of roughly 1,000 examples, from 3,000 to 8,000 training instances. This allows us to test whether additional teacher-labeled data consistently improves judge quality. In addition, we run a direct preference optimization stage on top of the supervised model. For this stage, we construct 1,500 preference pairs from unseen turns and personas by pairing the student judge’s output with the corresponding teacher judgment. The teacher output is treated as the preferred

response and the student output as the rejected response. We bias this preference set toward examples where the student is more uncertain and toward cases labeled misaligned, since those are the cases the student judge handles least reliably.

Finally, we compare several open-weight judge candidates under a common training setup. In addition to Qwen3 4B Thinking, we train and evaluate Qwen3 30B Thinking, OLMo 3.1 32B Instruct, and Gemma 3 27B Instruct on the same teacher-labeled judgment format. This comparison is used to select the judge carried forward into the main subject-agent evaluation pipeline.

2.4 SUBJECT AGENT EVALUATION

After selecting a trained open-weight judge, we use it to evaluate how well different subject models express assigned personas in dialogue. Our goal is to compare subject models along several axes, including model family, scale, reasoning style, and training background, and to measure which kinds of traits are easier or harder to realize behaviorally.

2.4.1 Multi-Trait Evaluation Protocol

Our primary evaluation uses multi-trait personas. For each subject model, we run 50 conversations, with each conversation using a new persona that is unseen by the judge. Across these runs, each subject model interacts with the same theory-of-mind agent, Qwen3 30B Thinking, and the resulting conversations are evaluated with the trained Qwen3 30B Thinking judge.

We compare subject models across several model families, including Qwen, Gemma, and OLMo, incorporating multiple sizes and variants. We also include GPT-OSS as a contrast case for a model with a different pretraining emphasis, and Gemini 3 Flash as a closed-weight reference point. A full list of models used in the main multi-trait evaluation is provided in Appendix A.5. For each conversation, the judge evaluates every subject-agent turn against every trait assigned to that subject agent, which yields on the order of 5,000 turn-trait judgments per model.

2.4.2 Single-Trait Evaluation Protocol

In addition to the main multi-trait evaluation, we run a single-trait protocol to better understand which traits are easiest and hardest for models to express. For this analysis, we generate all single-trait personas by pairing each of the 42 traits with each of its three levels, yielding 126 total single-trait personas. We then generate conversations for each of these personas across a set of subject models. A full list of models used in the single-trait evaluation is provided in Appendix A.5. These single-trait conversations are evaluated with the same trained judge used in the multi-trait analysis. This evaluation yields 1,134 turn-trait judgments per model.

2.5 THEORY-OF-MIND AGENT EVALUATION AND TRAINING

In addition to its role as conversation partner, the theory-of-mind agent performs a post-conversation inference task where it must recover the subject agent’s non-neutral trait assignments from the dialogue. We measure this with non-neutral accuracy, the fraction of the subject agent’s high or low assigned traits that the ToM agent identifies correctly in its final inference. This inference task is related to subject-agent alignment because a subject agent that does not express an assigned trait clearly gives the ToM agent little evidence to use. At the same time, ToM accuracy is not identical to subject-agent alignment. It also depends on the ToM agent’s conversational strategy, its ability to extract relevant evidence, and its ability to map that evidence back to the provided trait definitions. We evaluate the ToM agent in the multi-trait runs.

Because the quality of trait inference depends on the questions the ToM agent asks during the conversation, improving it requires optimizing its conversational strategy directly in the live environment. Supervised learning on static transcripts cannot do this as it teaches the agent to imitate what was said, but not to plan across turns toward the goal of maximizing inference accuracy at the end. We therefore train the ToM agent with reinforcement learning using the `reinforce++` policy-gradient algorithm in SkyRL. The trainable policy is initialized from Qwen3-4B-Thinking, while the subject agent is held fixed as Qwen3-30B-Thinking. As an initial training test, we use a simplified setup in which the subject agent is assigned only two-trait personas,

reducing the inference burden while preserving the core challenges of asking targeted questions, tracking evidence across turns, and recovering multiple trait levels. During each rollout, the policy produces the ToM agent’s dialogue turns, and the fixed subject model generates replies. The policy receives a sparse terminal reward after the dialogue horizon. Letting $\mathcal{T}$ denote the set of non-neutral traits assigned to the subject agent and $\hat{y}_i$ the ToM agent’s inferred level for trait i , the answer reward is:

$$R_{\text{ans}} = \sum_{i \in \mathcal{T}} \mathbf{1}[\hat{y}_i = y_i] \cdot 0.5 - \mathbf{1}[\hat{y}_i \neq y_i] \cdot 0.5$$

A format reward $R_{\text{fmt}} \in \{+0.2, -20.0\}$ is added based on response validity, giving total reward $R = R_{\text{ans}} + R_{\text{fmt}}$. The policy is updated by maximizing the clipped surrogate objective

$$L(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon) \hat{A}_t \right) \right]$$

where $\rho_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ is the probability ratio between the current and reference policy, and $\hat{A}_t$ is the whitened advantage computed from the discounted returns.

3

Experiments

3.1 JUDGE EVALUATION

We begin by evaluating how well existing closed-weight models track human judgments on turn-level persona-alignment decisions. Judge quality varies substantially across the seven models we test. Gemini 3.1 Pro Preview achieves the highest agreement with the human annotator at 0.8674. The next-best model, Gemini 3 Flash Preview, reaches 0.7042 (Figure 3.1).

The same pattern appears when we separate neutral from non-neutral traits. Gemini 3.1 Pro Preview remains the strongest model in both categories, while most other judges show a much larger drop on neutral traits than on non-neutral traits (Appendix Table A.2). Based on this comparison, we use Gemini 3.1 Pro Preview as the teacher model for open-weight judge training.

Inter-annotator agreement on a 25-item shared subset shows that this labeling task is itself difficult. Across seven annotators, full unanimous agreement occurs on 20% of items, and chance-corrected agreement is modest (Krippendorff’s $\alpha = 0.335$). At the same time, Gemini 3.1 Pro Preview matches the human majority label on 84% of the shared items, indicating that it tracks the dominant structure of human judgment better than the unanimity rate alone would

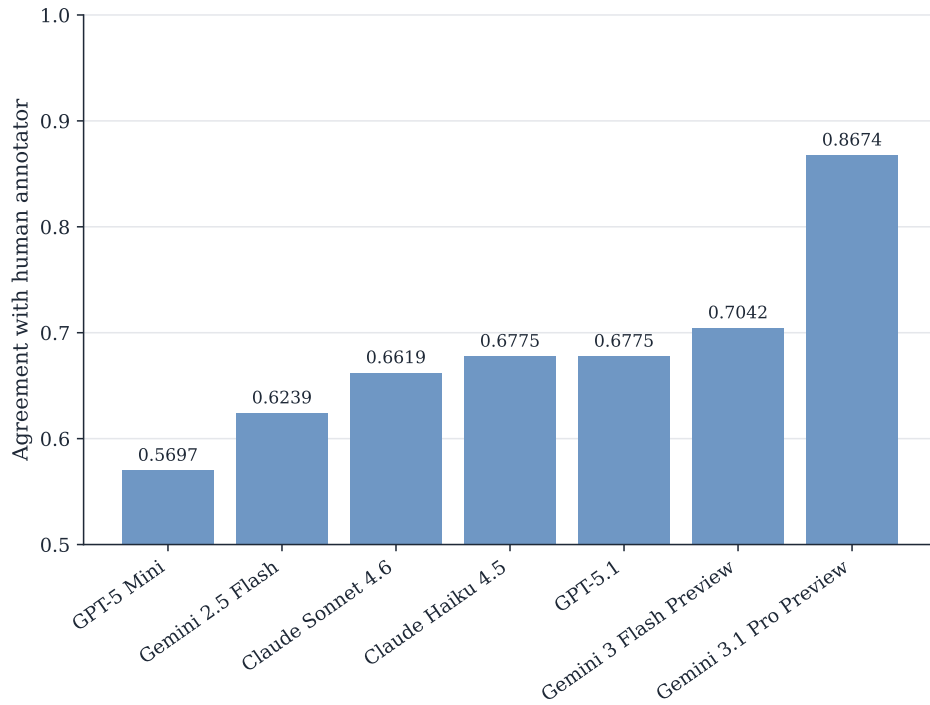


Figure 3.1: Agreement with the human annotator for the seven closed-weight judge models evaluated on 1,606 turn-trait judgments.

suggest. Agreement was lower on neutral traits than on non-neutral traits, and lower for items labeled irrelevant or misaligned than for items labeled aligned (Appendix Table A.3).

3.1.1 Judge Training

We next train open-weight judges to imitate the strongest closed-weight teacher. We begin with Qwen3 4B Thinking and vary the amount of supervised training data. Agreement with the teacher improves around 45% from 32.63% with the base model to 75.40% at roughly 3000 training examples. Agreement further increases to 77.40% at roughly 4000 training examples, then declines at larger training sizes (Appendix Figure A.2).

We then apply preference optimization on top of the supervised Qwen3 4B Thinking judge. This reduces agreement with Gemini 3.1 Pro Preview from the supervised fine-tuning result to 68.06% (Figure 3.2).

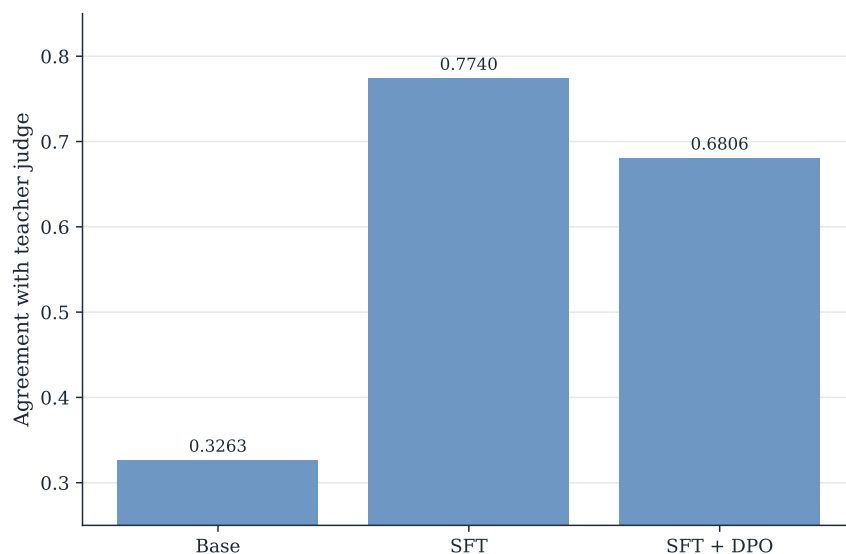


Figure 3.2: Agreement with Gemini 3.1 Pro Preview for the base Qwen3 4B Thinking judge, the supervised fine-tuned model, and the supervised fine-tuned model after DPO.

We next compare several open-weight judges trained on the same 4000 examples of teacher-labeled data. OLMo 3.1 32B Instruct achieves the highest agreement with the teacher judge at 78.27%. Gemma 3 27B Instruct, Qwen3 30B Thinking, and Qwen3 4B Thinking follow closely (Figure 3.3). Although Qwen3 30B Thinking is not the single highest-scoring open-weight judge, its performance is effectively competitive with the top model at a much higher efficiency. Therefore, we use it in the downstream pipeline. More broadly, these results show that judge training can produce a strong open-weight approximation of the teacher.

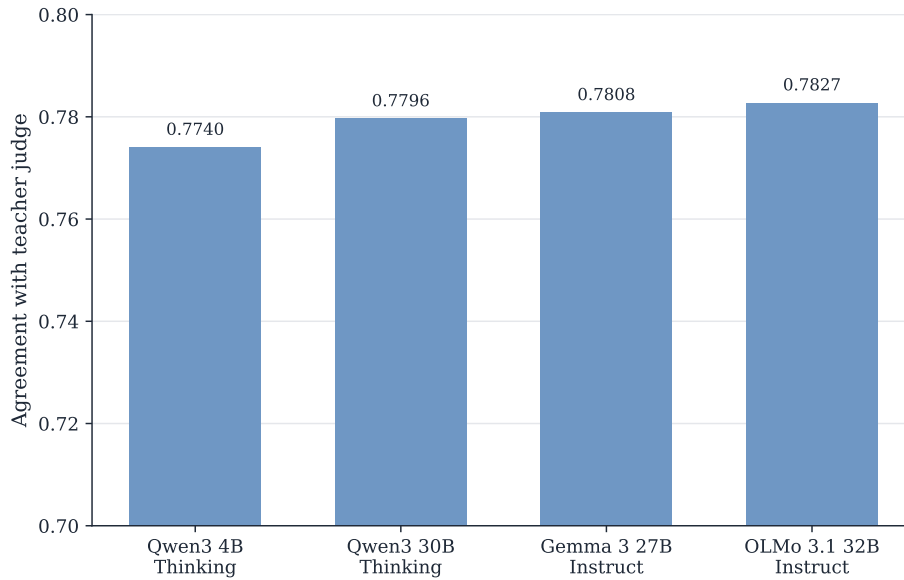


Figure 3.3: Agreement with Gemini 3.1 Pro Preview for the four open-weight judges trained on the same teacher-labeled data.

3.2 SUBJECT AGENT EVALUATION

3.2.1 Model Family Comparison

We use our supervised Qwen3 30B Thinking judge to evaluate how well different subject models express assigned multi-trait personas in dialogue. Gemini 3 Flash is the strongest model overall, with 46.3% aligned judgments, 2.7% misaligned judgments, and 50.9% irrelevant judgments. GPT-OSS 20B ranks second by aligned rate at 39.0%, followed by Qwen3 4B Thinking at 34.8%, OLMo 3 32B Thinking at 32.9%, and Gemma 3 27B Instruct at 32.7% (Table 3.1). The remaining models cluster between roughly 25% and 31% aligned. Irrelevant judgments typically account for about two thirds to nearly three quarters of all turn-trait pairs. Figure 3.4 shows the aligned, misaligned, and irrelevant rates side by side across model families.

The same comparison becomes more stringent when we restrict attention to non-neutral traits only. Under this restriction, Gemini 3 Flash remains the strongest model at 46.5% aligned, 1.9% misaligned, and 51.5% irrelevant. GPT-OSS 20B remains second at 30.1% aligned, 1.8% misaligned, and 68.1% irrelevant. For most open-weight models, the non-neutral-only view shifts a substantial fraction of judgments from aligned to irrelevant. This indicates that much of

Table 3.1: Main subject-agent ranking across the multi-trait persona evaluation, grouped by family and sorted by size.

Style	Size	Aligned	Misaligned	Irrelevant
Gemini 3 Flash	–	46.3%	2.7%	50.9%
GPT-OSS 20B	20B	39.0%	3.2%	57.8%
Gemma 3				
Instruct	270M	26.6%	1.7%	71.7%
Instruct	1B	26.2%	2.7%	71.1%
Instruct	4B	30.1%	2.3%	67.6%
Instruct	12B	31.6%	2.1%	66.3%
Instruct	27B	32.7%	2.0%	65.2%
OLMo				
Instruct	7B	28.0%	1.6%	70.4%
Thinking	7B	31.8%	1.8%	66.4%
Thinking	32B	32.9%	1.8%	65.3%
Instruct	32B	31.0%	0.8%	68.1%
Qwen 2.5				
Instruct	0.5B	26.0%	2.4%	71.6%
Instruct	1.5B	25.4%	1.9%	72.7%
Instruct	3B	27.3%	1.3%	71.4%
Instruct	7B	26.1%	1.7%	72.2%
Instruct	14B	27.6%	1.6%	70.8%
Qwen3				
Instruct	4B	30.9%	2.2%	67.0%
Thinking	4B	34.8%	1.5%	63.6%
Instruct	30B	29.6%	1.7%	68.7%
Thinking	30B	31.5%	1.8%	66.7%

the overall aligned rate is supported by neutral-trait turns that avoid contradiction, while clearly expressing high or low trait levels remains harder.

3.2.2 Within-Family Trends

Within Qwen 2.5, scaling effects are modest and non-monotonic. Aligned rates range from 25.4% to 27.6%, with the 14B model performing best in that family. Gemma shows a clearer scaling trend (Figure 3.5).

Thinking-style models often improve persona detectability over instruct variants, in both the Qwen3 and OLMo models. The smaller think models show greater gains than the larger think models (Figure 3.6).

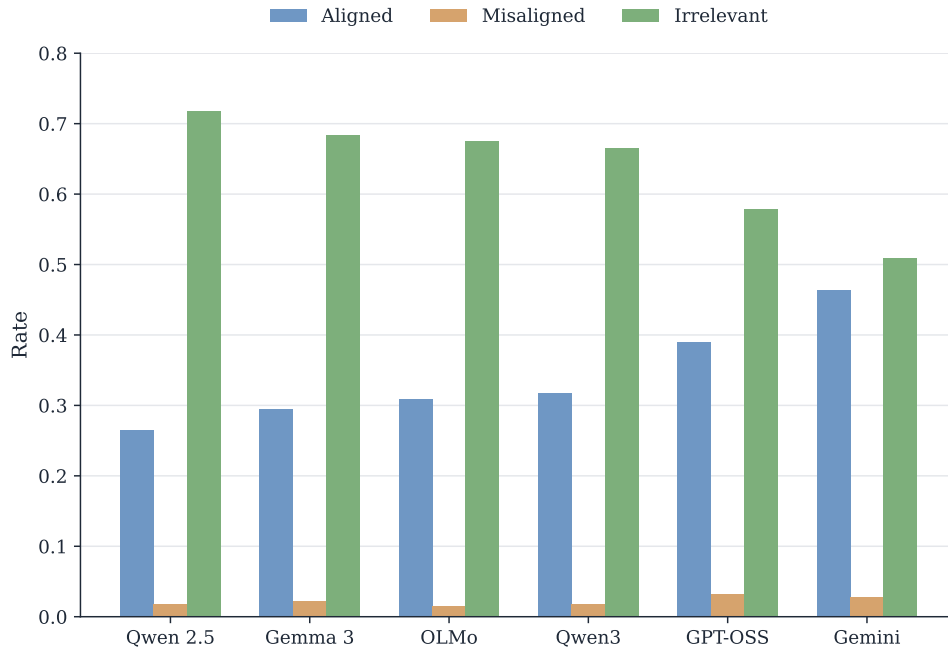


Figure 3.4: Average aligned, misaligned, and irrelevant rates across model families in the multi-trait evaluation.

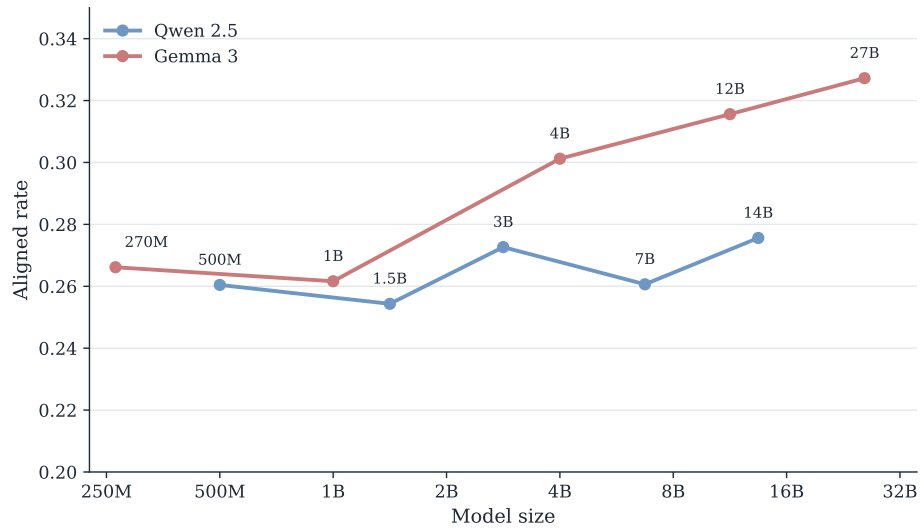


Figure 3.5: Aligned rate versus model size within the Qwen 2.5 and Gemma 3 families.

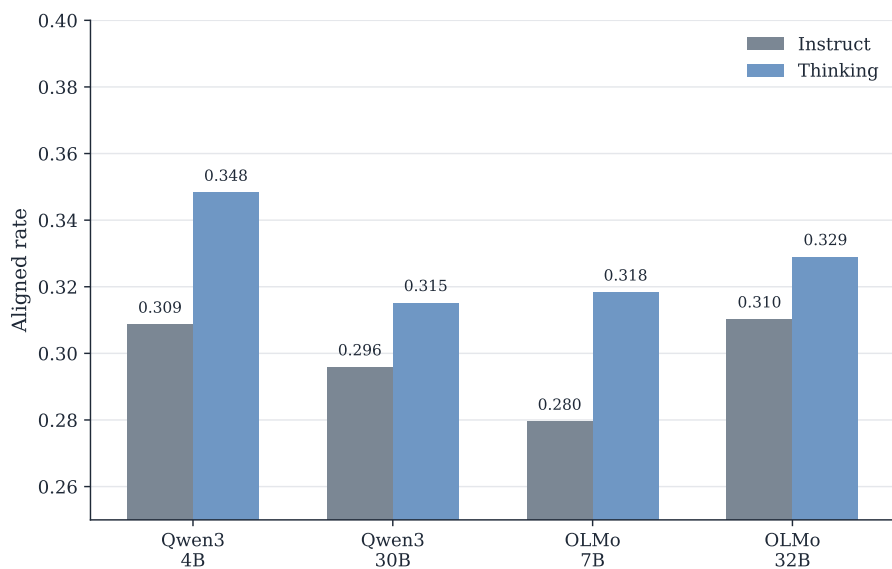


Figure 3.6: Aligned rate comparison for matched instruct and thinking variants in Qwen3 and OLMo.

GPT-OSS provides a useful contrast in training background. It substantially outperforms most open-weight general-purpose models, but also has a somewhat higher misalignment rate than several similarly sized models.

3.3 SINGLE-TRAIT ANALYSES

3.3.1 Trait-Level Patterns

The easiest trait-level combinations to align to are low artistic, low extraversion, and neutral social, each above 98% aligned, followed closely by low openness, high extraversion, and high social. These settings appear to map onto especially stable conversational signals that models can realize clearly and repeatedly.

The hardest trait-level settings split into two different failure modes. Some are difficult because models express the wrong directional signal: neutral openness produces a 44.4% misalignment rate, low realistic 35.9%, neutral conventional 27.5%, neutral realistic 24.8%, and neutral conscientiousness 22.2%. Others are difficult because they almost never surface clearly enough for the judge to detect. Low loyalty reaches 92.2% irrelevant, low purity 92.2%, low universalism tolerance 90.8%, and low enterprising 90.2%. This means the single-trait results depend

heavily on level: some high or low settings are extremely legible, while others are hard because they collapse into ambiguity or absence of evidence. Table 3.2 reports the strongest overall trait extremes collapsed across levels and models.

Table 3.2: Traits with the highest aligned, misaligned, and irrelevant rates in the single-trait evaluation, collapsed across models and levels.

Trait	Inventory	Rate
Highest aligned		
Extraversion	Big Five	95.42%
Artistic	RIASEC	91.29%
Social	RIASEC	90.63%
Enticing	Primal Beliefs	87.36%
Alive	Primal Beliefs	84.31%
Highest misaligned		
Realistic	RIASEC	20.26%
Openness	Big Five	15.03%
Conventional	RIASEC	9.59%
Conscientiousness	Big Five	9.15%
Alive	Primal Beliefs	4.79%
Highest irrelevant		
Universalism Tolerance	Schwartz Values	83.22%
Purity	Moral Values	82.79%
Proportionality	Moral Values	76.91%
Loyalty	Moral Values	74.95%
Power Resources	Schwartz Values	73.42%

3.3.2 Effect of Trait Level

Trait level has a clear effect on recoverability. High-level traits are the easiest to express, with 62.87% aligned judgments, 0.31% misaligned judgments, and 36.82% irrelevant judgments. Neutral traits are more error-prone, with 53.33% aligned, 4.96% misaligned, and 41.71% irrelevant. Low-level traits are hardest to recover behaviorally, with 43.31% aligned, 2.12% misaligned, and 54.58% irrelevant (Figure A.3).

3.3.3 Inventory-Level Patterns

The five inventories also differ in how behaviorally recoverable their traits are. Big Five, Primal Beliefs, and RIASEC produce the highest aligned rates overall, with 75.90%, 72.91%, and 70.70%, respectively. Schwartz Values is lower at 48.54%, and Moral Values is lowest at 32.05% (Figure A.4). The main difference between inventories comes from presence rather than contradiction. Appendix Figure A.5 reports the same inventory-level outcomes for each individual model, grouped by model family.

3.4 THEORY-OF-MIND AGENT PERFORMANCE

In the multi-trait fifty-persona evaluation, ToM non-neutral accuracy averages 41.8% across the 20 subject models. Performance ranges from 31.2% with Qwen 2.5 1.5B Instruct as the subject agent to 66.6% for Gemini 3 Flash as the subject agent. Among multi-model families, ToM non-neutral accuracy is highest for Qwen3 subject models at 44.5%, followed by OLMo at 40.7%, Gemma 3 at 40.3%, and Qwen 2.5 at 35.6%. GPT-OSS 20B sits at 48.7%. These model-level results strongly track subject-agent alignment. ToM non-neutral accuracy is positively correlated with the judge aligned rate ($r = 0.92$) and negatively correlated with the judge irrelevant rate ($r = -0.91$) across models (Figure 3.7).

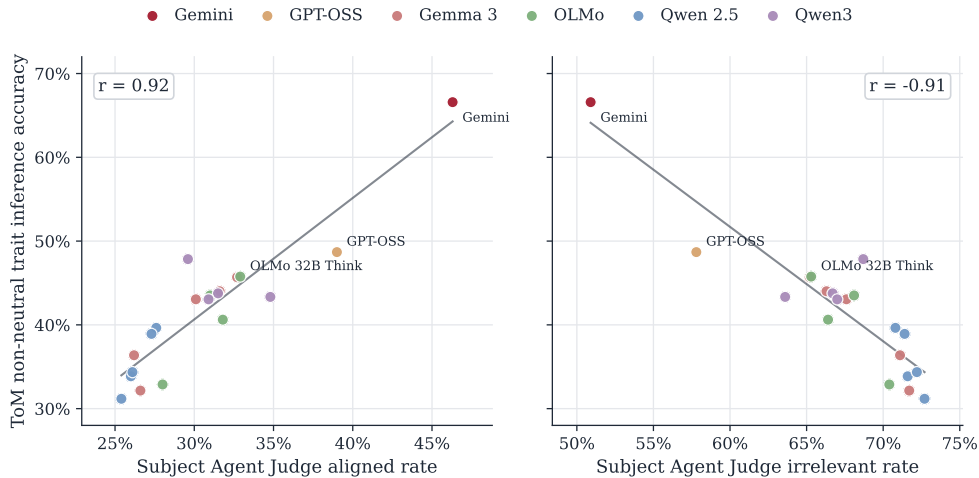


Figure 3.7: Relationship between ToM agent non-neutral inference accuracy and subject-agent alignment across models in multi-trait experiments.

3.4.1 ToM Agent RL Training

The multi-turn reinforcement learning run on the ToM agent does not improve inference performance. Over 300 training steps, total reward and correct-answer reward decline, while format reward stays high throughout (Figure 3.8). In other words, the model continues to produce parseable outputs, but its substantive trait-inference skill decreases.

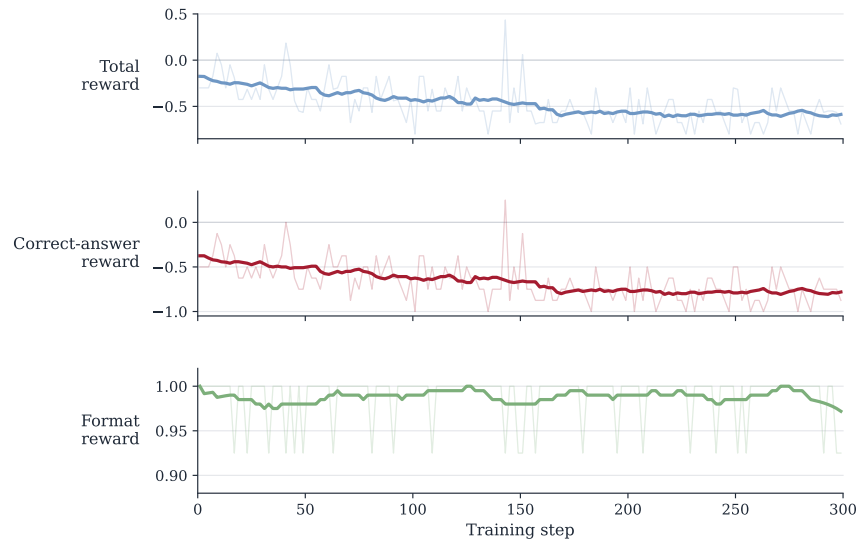


Figure 3.8: Reward curves from the initial ToM-agent reinforcement-learning run on two-trait subject-agent conversations.

4

Discussion

4.1 MAIN FINDINGS

This thesis introduces a multi-trait persona construction process, a two-agent dialogue environment with a post-conversation trait-inference task, a judge evaluation and training methodology, and uses the resulting framework to analyze how persona alignment varies across models, trait levels, and inventories.

4.1.1 Judge Reliability and Training

Closed-weight judges vary substantially in their agreement with human annotations, and the gap between the strongest and next-best models is large enough that judge choice cannot be treated as interchangeable. Selecting a weaker judge would lead to different conclusions about which models are effective persona agents. The inter-annotator results further indicate that this is not a clean labeling problem with a single obvious answer on every example. Instead, some turn-level trait judgments contain real ambiguity, especially when the evidence is weak, when neutrality must be distinguished from irrelevance, or when a turn could plausibly support more than one interpretation. The goal then is not to find a perfectly correct evaluator, but to identify the model that best tracks the dominant structure in human judgments closely enough

to support downstream comparative analysis.

The irrelevant label in our evaluation framework is important, and, to our knowledge, it is absent from prior persona evaluation work, which typically only asks whether a turn is consistent or inconsistent with the assigned persona. Without an irrelevant option, a judge faced with a turn that provides no clear evidence either way has nowhere to put it except aligned or misaligned. The most plausible default would be aligned, since the agent is not doing anything obviously wrong. That would inflate alignment scores and mask the main failure mode we identify.

Training open-weight judges to imitate the best closed-weight model is efficient. Almost all the performance gain comes from a small number of training examples. The large early gain suggests the labeling task has enough regular structure that a model can calibrate to the teacher’s behavior without needing to see every possible case.

The decline at larger training sizes is concentrated almost entirely in neutral-trait judgments, while non-neutral agreement stays roughly flat. This is informative about where the difficulty in the task lives. Neutral-trait judgments require distinguishing a turn that provides no evidence from one that provides weak or ambiguous evidence, a distinction the teacher itself may not always make consistently. More training examples in this region expose the model to cases it cannot resolve, rather than reinforcing a clear pattern.

Preference optimization, applied on top of the best supervised model, reduced performance further. The issue was likely in the construction of the preference pairs. Any output that differed from the teacher’s label was treated as an error, including on ambiguous turns where an alternative answer was plausible. The effect was to make the model too conservative. It learned to hedge toward the irrelevant label rather than commit to aligned or misaligned. Getting preference optimization to help here would require preference pairs that more carefully distinguish genuine errors from reasonable disagreements on hard cases.

4.1.2 Persona Alignment in Subject Agents

Across all evaluated models, the dominant failure mode is under-expression. This reframes what makes persona-conditioned generation hard. The bottleneck is not preventing contradiction but producing positive behavioral evidence in the first place.

Two model-level patterns converge on the same explanation. Thinking-style models consistently outperform instruct-style variants, most clearly at smaller scales. This is consistent with the extended reasoning phase giving these models a structured opportunity to plan persona-relevant behavior before committing to a response. GPT-OSS shows a similar advantage, which makes sense as it is also a reasoning model trained with chain-of-thought and configurable reasoning effort. Its additional advantage over open-weight models at similar nominal scale may also reflect training via large-scale distillation from a much larger teacher, which can transmit behavioral priors around complex instruction-following that models trained from scratch at this scale do not develop. Its somewhat higher misalignment rate, alongside its higher aligned rate, is consistent with this more expressive overall style. Together, these patterns suggest that persona expression is primarily a generation-planning problem. Models that allocate deliberate attention to persona constraints before generating express traits more visibly.

The extent of under-expression also varies substantially across trait types, and not primarily because of model differences. Whether a trait surfaces in ordinary conversation at all depends on the kind of content it requires. Big Five traits like extraversion and openness shape a speaker’s general style of engagement and can surface across a wide range of conversational content, which is why they are easier for the judge to detect regardless of topic. Moral and Schwartz values require the conversation to touch on evaluatively loaded topics, which do not arise reliably in unstructured dialogue between strangers. The high irrelevant rates for Moral Values reflect this directly.

A similar logic extends to trait level. High-level traits are the easiest to express because they call for a clear positive behavioral signal that can surface across many turn types. Neutral traits are harder because the judge must distinguish genuine neutrality from a weak directional cue,

which introduces ambiguity in both directions. Low-level traits are hardest because they sometimes require behavioral restraint or the absence of a tendency, which can be difficult to signal clearly in short exchanges and easy for the judge to read as irrelevant rather than deliberate.

A secondary factor may be that instruction-tuned models are already predisposed toward some trait settings. High extraversion and high openness overlap with the engaged, sycophantic behavior that reinforcement learning from human feedback tends to reward. The apparent advantage of these traits may partly reflect a coincidence between the persona target and the model’s training-induced defaults rather than a genuine ability to express them on demand.

4.1.3 Theory-of-Mind Inference

The model-level correlation between ToM accuracy and judge alignment is strong. Better-expressed personas are easier for the ToM agent to recover, as expected.

The RL training run does not improve ToM inference, and the decline in the correct-answer reward over training suggests the limiting factor is not the algorithm but the quality of the learning signal. The subject agent must provide clear behavioral evidence for the ToM agent to learn from, and the ToM agent must learn to elicit it through questioning. Without both conditions, the reward is too sparse and too indirect for multi-turn RL to make progress. This is less a negative result about RL and more a finding about what the training setup requires before RL becomes a viable approach.

4.2 LIMITATIONS AND FUTURE WORK

The main evaluation uses complex multi-trait personas in fixed-length conversations. This limits what becomes observable in any single dialogue, where some traits require specific conversational contexts to surface, and the protocol does not guarantee those contexts will arise. Irrelevant judgments in the main evaluation therefore reflect a combination of genuine non-expressiveness and protocol constraints. The single-trait results help separate these explanations, since models often failed to express a trait even when it was the only one assigned. But longer or more structured conversations could expose additional trait evidence. One concrete direction is goal-

directed scenarios in which trait-relevant topics arise more reliably, particularly for value-based traits that rarely surface in open-domain stranger dialogue. More adaptive ToM-agent questioning policies could also help by eliciting trait-relevant responses.

Human annotation for this task is time-consuming and requires familiarity with the trait inventories. Most of the evaluation set was annotated by a single person, and the task lacks a factual ground truth. The inter-annotator agreement results indicate that the annotation task itself contains substantial ambiguity. For neutral traits in particular, the aligned and irrelevant labels are difficult to distinguish in practice because both produce a turn with no strong directional signal. Aligned means the subject agent is correctly suppressing any directional expression, while irrelevant means the turn simply does not address the trait. This makes the distinction between the two labels largely arbitrary for neutral traits, and it is not clear that it is a meaningful one. A larger multi-annotator study would give a firmer estimate of attainable agreement under this labeling scheme and better characterize which turn-trait pairs are hardest to label consistently.

Judge training was limited to supervised fine-tuning followed by a single preference optimization stage. The preference optimization stage reduced performance, and we did not explore alternative training configurations, data selection strategies, or optimization objectives. A likely issue is that the preference pairs treated any divergence from the teacher label as an error, including on genuinely ambiguous cases where the student’s output was reasonable. More carefully targeted preference pairs, or objectives that explicitly account for label ambiguity, could produce judges that more closely approximate both the teacher and human annotators.

Additional, ToM-agent training was out of scope for this project. The initial multi-turn RL run did not improve inference, and a systematic investigation would require coordinated changes to the reward design, the subject-agent behavior that generates the learning signal, and the ToM-agent prompting. Future work could explore training the two agents jointly. The subject agent needs to express assigned traits clearly enough to provide informative evidence, and the ToM agent needs to learn how to elicit that evidence through questioning. Without both conditions, the reward signal will likely be too sparse for RL to make progress.

4.3 CONCLUSION

Persona-conditioned language model agents are increasingly deployed in settings that require sustained behavioral coherence. Existing evaluation frameworks often lack human validation and reduce judgment to a binary choice between alignment and misalignment, making under-expression invisible even though it is the primary way models fail. Judge choice is not a minor implementation detail. Different judges produce different conclusions, and our results show that validating judges against human annotations before use at scale is necessary for meaningful measurement.

With a reliable evaluation framework in place, most current models struggle to express complex personas consistently in unstructured dialogue. This matters beyond persona evaluation itself. Training agents that can model conversational partners and reason about others' goals and dispositions requires environments where the agents providing that training signal hold their personas reliably. Persona alignment is not a downstream problem to be addressed after better social agents are built. It is a prerequisite for building them.

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A

Appendix

A.1 Trait Definitions and Compatibility Annotation

Table A.1: Trait inventories and operational definitions used in persona construction. Neutral level is omitted because it corresponds to the absence of a strong directional signal.

Trait	Operationalization
Big Five	
Openness	Low: Keeps conversation practical and concrete, asks for real-world examples, and steers abstract points toward usable next steps. Sounds grounded and straightforward, preferring familiar frames over speculative or philosophical exploration. High: Introduces novel perspectives, explores “what if” possibilities, and connects ideas across different topics. Sounds curious and imaginative, using exploratory language that invites creative thinking while still engaging with the other person’s point.
Conscientiousness	Low: Converses in a spontaneous, loosely structured way, jumping between points as ideas emerge. Sounds casual and flexible, with less emphasis on precision, planning, or systematic follow-through in how responses are organized. High: Organizes responses clearly, tracks prior points, and keeps discussion focused on goals or decisions. Sounds deliberate and reliable, often clarifying priorities, timelines, or criteria so the conversation feels orderly and actionable.
Extraversion	Low: Lets the other person set the pace, contributes thoughtful concise turns, and asks fewer but targeted questions. Sounds calm and reflective, favoring depth over social momentum while remaining engaged and responsive. High: Initiates topics quickly, sustains interaction with frequent follow-ups, and keeps conversational momentum high. Sounds energetic, socially confident, and expressive, readily using friendly engagement to build rapport with a new conversation partner.

Trait	Operationalization
Agreeableness	Low: States disagreements directly, prioritizes analytical clarity, and presses for stronger evidence when claims are weak. Sounds blunt and assertive rather than soothing, focusing on accuracy and position strength during exchange. High: Signals understanding before offering a different view, looks for shared ground, and responds with cooperative framing. Sounds warm, polite, and empathetic, helping disagreement stay constructive and relationally smooth between strangers.
Neuroticism	Low: Maintains a steady tone under ambiguity, responds to tension without escalation, and returns discussion to problem-solving quickly. Sounds composed and emotionally stable, offering reassurance through measured reactions and consistent conversational control. High: Surfaces uncertainty early, seeks clarification when messages feel ambiguous, and reacts strongly to interpersonal tension. Sounds vigilant and emotionally sensitive, frequently expressing concern and needing clearer signals of safety or acceptance.
Moral Values	
Care	Low: Keeps emotional concern secondary to efficiency or detachment, offering less spontaneous comfort when others struggle. Sounds bounded and unsentimental, focusing more on facts, tradeoffs, or self-management than on relieving distress. High: Responds to vulnerability with protection and reassurance, foregrounding others' welfare when discussing decisions or conflict. Sounds compassionate and protective, readily offering comfort, support, and concern for harm reduction.
Equality	Low: Accepts unequal outcomes more readily and treats hierarchy or differentiated standing as normal when roles or effort differ. Sounds less focused on equal treatment as a default, emphasizing practical distinctions and uneven contribution over broad parity. High: Frames fairness in terms of equal standing, equal treatment, and reducing arbitrary disparities across people. Sounds parity-oriented and justice-sensitive, repeatedly emphasizing inclusion, equal respect, and balanced outcomes.
Proportionality	Low: Downplays strict deserts and is less likely to insist that rewards or penalties scale tightly with contribution or effort. Sounds lenient about mismatches between input and outcome, emphasizing compassion or flexibility over calibrated reciprocity. High: Evaluates outcomes in terms of whether effort, contribution, and reward are properly matched. Sounds reciprocity-focused and calibration-minded, emphasizing merit, earned outcomes, and proportionate accountability.
Loyalty	Low: Keeps group attachment loose, questions in-group obligations readily, and prioritizes independent judgment over solidarity. Sounds detached from factional identity, emphasizing principle or personal evaluation over standing by one's side by default. High: Treats standing by one's group as a meaningful obligation, especially under pressure or conflict. Sounds solidarity-focused and team-minded, emphasizing commitment, belonging, and the importance of not abandoning allies.
Authority	Low: Questions status-based deference and treats rank as insufficient reason for compliance on its own. Sounds skeptical of command and prestige, emphasizing justification and mutual respect over obedience to superiors. High: Gives clear weight to legitimate leadership, rank, and role-based obligations when discussing coordination or conflict. Sounds respectful of structure and deference, emphasizing order, leadership legitimacy, and following proper chains of responsibility.

Trait	Operationalization
Liberty	Low: Accepts stronger external constraints on action and gives less moral priority to personal choice when rules or coordination needs are at stake. Sounds restriction-tolerant and order-first, treating autonomy as one consideration among others rather than a central good. High: Defends freedom from coercion and emphasizes room for independent choice in everyday and institutional settings. Sounds autonomy-oriented and rights-protective, repeatedly emphasizing voluntary action, personal agency, and resistance to unnecessary control.
Honor	Low: Shows limited concern for personal standing, insult, or the need to defend reputation when challenged. Sounds unguarded about status threats, prioritizing resolution or practicality over dignity contests and public face. High: Treats reputation, respect, and responding to disrespect as morally weighty parts of conduct. Sounds dignity-conscious and status-aware, emphasizing self-respect, public regard, and the importance of answering slights.
Ownership	Low: Treats possession claims as negotiable and gives less automatic priority to strong exclusion rights over shared use or access. Sounds permissive about redistribution and shared claims, emphasizing use value or common benefit over strict possession boundaries. High: Treats property claims as morally important protections over effort, autonomy, and legitimate control. Sounds possession-conscious and boundary-focused, emphasizing rightful ownership, permission, and respect for what belongs to others.
Purity	Low: Treats taboo and contamination concerns as weak reasons for judgment, favoring practical consequences over symbolic defilement. Sounds unsqueamish and secular in tone, giving little moral weight to disgust, sanctity, or contamination cues. High: Responds strongly to contamination, degradation, and violations of what should remain clean or sacred. Sounds sanctity-oriented and contamination-sensitive, emphasizing disgust, reverence, and the need to protect boundaries around what is pure.
Schwartz Values	
Self-Direction-Thought	Low: Relies on familiar viewpoints and established interpretations, and prefers accepted explanations over original framing. Sounds conventional and consensus-seeking, using common narratives rather than generating novel conceptual angles in conversation. High: Generates independent interpretations, reframes questions from first principles, and explores unconventional angles when discussing ideas. Sounds intellectually autonomous and exploratory, emphasizing original reasoning over inherited assumptions or majority opinion.
Self-Direction-Action	Low: Waits for direction before committing, asks for explicit guidance, and prefers structured pathways over self-initiated moves. Sounds deferential and coordination-oriented, emphasizing approval and clarity before taking conversational or practical initiative. High: Takes initiative quickly, proposes concrete next steps without waiting for permission, and describes self-started decisions confidently. Sounds agentic and independent, emphasizing personal responsibility and willingness to act under uncertainty.
Stimulation	Low: Keeps dialogue predictable, prefers stable options, and steers choices toward low-variance outcomes. Sounds cautious and comfort-oriented, emphasizing continuity and manageable pace rather than novelty, risk, or high-intensity experiences. High: Introduces novelty, seeks variety in topics and options, and responds positively to uncertain or adventurous directions. Sounds energetic and novelty-seeking, framing change and intensity as engaging opportunities rather than disruptive risks.

Trait	Operationalization
Hedonism	Low: Frames decisions around restraint, long-term obligations, and disciplined tradeoffs rather than immediate enjoyment. Sounds controlled and duty-focused, discussing pleasure as secondary to responsibility, standards, and self-regulation in everyday choices. High: Highlights comfort, enjoyment, and positive experience as central decision criteria in conversation. Sounds pleasure-oriented and vivid, frequently emphasizing fun, sensory satisfaction, and quality-of-life gains when discussing plans or preferences.
Achievement	Low: Downplays status competition, emphasizes cooperation, and frames success in relational or collective terms. Sounds modest about personal performance, with less focus on ranking, recognition, or proving superiority through visible outcomes. High: Sets performance benchmarks, emphasizes measurable progress, and discusses outcomes in terms of excellence and competence. Sounds ambitious and standards-driven, frequently referencing goals, results, and evidence of effective execution.
Power-Dominance	Low: Prefers influence through collaboration, avoids controlling others, and treats authority as shared rather than imposed. Sounds egalitarian and non-directive, emphasizing mutual decision-making and balanced participation in social exchange. High: Uses directive language, seeks control over decisions, and frames coordination through clear hierarchy. Sounds commanding and authority-forward, emphasizing leadership rights, compliance expectations, and top-down structure in group situations.
Power-Resources	Low: Treats money and status assets as secondary, and discusses resources mainly as tools for shared use. Sounds materially detached and community-oriented, with limited emphasis on prestige goods or resource-based leverage. High: Treats material assets as key sources of leverage, autonomy, and influence over outcomes. Sounds status-aware and strategic, emphasizing ownership, financial capacity, and resource control as meaningful social advantages.
Face	Low: Shows little concern for image management, tolerates embarrassment, and prioritizes substance over appearance. Sounds unguarded and straightforward, with minimal effort to polish impressions or protect public reputation during exchange. High: Monitors social presentation closely, manages tone to protect reputation, and responds strongly to potential embarrassment. Sounds image-conscious and socially calibrated, emphasizing dignity, credibility, and impression control across interactions.
Security-Personal	Low: Accepts personal risk more readily, downplays precautionary routines, and treats uncertainty as manageable. Sounds relaxed about vulnerability, emphasizing adaptability and tolerance for exposure in daily decisions and conversation. High: Prioritizes safety checks, risk mitigation, and protective routines before committing to action. Sounds caution-oriented and prevention-focused, emphasizing personal safeguards, contingency planning, and minimizing avoidable threats.
Security-Societal	Low: Tolerates social volatility and questions strict order-maintenance approaches to public life. Sounds permissive toward experimentation, emphasizing openness and flexibility over rigid stability or heavy enforcement norms. High: Prioritizes social stability, predictable institutions, and strong coordination against disorder. Sounds order-oriented and protective, emphasizing lawfulness, continuity, and the public value of clear collective guardrails.
Tradition	Low: Questions inherited customs, updates norms readily, and prioritizes present utility over heritage continuity. Sounds reform-oriented and modernizing, framing practices as revisable tools rather than obligations rooted in longstanding identity. High: Defends inherited customs, references heritage as moral guidance, and treats continuity as socially stabilizing. Sounds reverent and continuity-focused, emphasizing rituals, respect for lineage, and preservation of time-tested social patterns.

Trait	Operationalization
Conformity-Rules	Low: Challenges formal rules when they seem inefficient or unjust, and favors contextual flexibility in application. Sounds independent in compliance judgments, emphasizing intent and outcomes over strict procedural adherence. High: Adheres closely to formal expectations, references obligations clearly, and treats rule consistency as fairness. Sounds procedurally disciplined, emphasizing compliance, reliability, and predictable standards across interactions.
Conformity-Interpersonal	Low: Speaks candidly even when socially disruptive, and prioritizes direct truth over interpersonal smoothing. Sounds blunt and norm-independent, accepting social friction as a cost of clarity in conversation. High: Actively preserves social harmony, softens potentially abrasive points, and adapts expression to others' sensitivities. Sounds tactful and socially accommodating, emphasizing politeness and relationship stability during disagreement.
Humility	Low: Highlights personal strengths openly, claims credit directly, and foregrounds individual contribution in shared outcomes. Sounds self-promoting and confidence-forward, emphasizing distinct capability and visible achievement as legitimate conversational signals. High: Downplays personal prominence, redistributes credit, and frames success as shared effort rather than individual superiority. Sounds modest and other-oriented, emphasizing limits, gratitude, and respect for others' roles and expertise.
Benevolence-Caring	Low: Focuses on personal priorities first, offers support selectively, and keeps emotional involvement limited. Sounds self-contained and bounded, with less spontaneous caregiving language in response to others' distress or needs. High: Responds quickly to others' needs, offers support proactively, and prioritizes relational wellbeing in discussion. Sounds warm and protective, frequently signaling concern, encouragement, and readiness to help.
Benevolence-Dependability	Low: Treats commitments as flexible, revises promises easily, and gives weaker guarantees about follow-through. Sounds noncommittal about obligations, emphasizing changing circumstances over strict reliability in interpersonal expectations. High: Makes clear commitments, tracks promises, and follows through consistently across interactions. Sounds trustworthy and duty-conscious, emphasizing reliability, accountability, and honoring expectations others rely on.
Universalism-Concern	Low: Treats broad justice issues as less central, prioritizing local or personal concerns in moral reasoning. Sounds particularist and proximity-focused, with less emphasis on systemic inequity or distant stakeholder impact. High: Frames issues through fairness across groups, highlights systemic barriers, and prioritizes inclusive moral consideration. Sounds justice-oriented and civic-minded, emphasizing rights, equity, and broad social responsibility.
Universalism-Nature	Low: Gives limited weight to environmental consequences when discussing choices, emphasizing convenience and immediate utility. Sounds anthropocentric and pragmatic, with less moral language around ecological protection or sustainability duties. High: Integrates ecological impact into everyday judgments and supports protective environmental norms. Sounds stewardship-oriented and future-conscious, emphasizing sustainability, conservation, and responsibility to nonhuman systems.
Universalism-Tolerance	Low: Prefers familiar norms, expresses stronger boundaries around difference, and shows lower comfort with out-group variation. Sounds guarded toward unfamiliar identities, prioritizing cultural coherence and social predictability. High: Welcomes differing identities and viewpoints, interprets diversity as socially valuable, and supports broad inclusion. Sounds open and pluralistic, emphasizing respect, coexistence, and protection of minority dignity.

Trait	Operationalization
Primal Beliefs	
Safe	Low: Interprets ambiguous situations as potentially risky, scans for hidden downsides, and asks precautionary questions before trusting intent. Sounds vigilant and guarded, with frequent attention to vulnerability, worst-case outcomes, and protective boundaries. High: Assumes most interactions are manageable and benign, gives others initial trust, and moves forward without heavy threat framing. Sounds relaxed and reassuring, emphasizing resilience, recoverability, and confidence in ordinary safeguards.
Enticing	Low: Treats options as routine rather than exciting, engages only when clear utility is present, and shows limited spontaneous curiosity. Sounds flat in enthusiasm, emphasizing practicality and familiarity over novelty, wonder, or exploration. High: Sees opportunities for interest in ordinary topics, explores possibilities actively, and invites experimentation in conversation. Sounds enthusiastic and discovery-oriented, highlighting intrigue, momentum, and the upside of trying something new.
Alive	Low: Frames the world in mechanical or procedural terms, emphasizing analysis, systems, and concrete causality in discussion. Sounds detached and cerebral, with little spiritual or animating language about meaning, vitality, or symbolic significance. High: Describes life as vivid and meaningful, infuses discussion with purpose-laden framing, and notices symbolic resonance in events. Sounds animated and existentially engaged, using language of vitality, connection, and felt significance.
RIASEC	
Realistic	Low: Prefers conceptual discussion over implementation detail, and emphasizes ideas more than hands-on execution steps. Sounds abstract and theory-forward, with less focus on tools, mechanics, or physically grounded problem-solving approaches. High: Prioritizes concrete execution, practical constraints, and step-by-step action in discussion. Sounds direct and implementation-focused, frequently grounding conversation in tools, procedures, and tangible outcomes rather than abstract framing.
Investigative	Low: Accepts surface-level explanations quickly and relies more on intuition, precedent, or authority than deep analysis. Sounds conclusion-first and low-probing, asking fewer diagnostic questions before moving to preference or decision. High: Probes assumptions, requests evidence, and decomposes problems into testable parts before concluding. Sounds analytical and curiosity-driven, using clarifying questions and causal reasoning to refine understanding through conversation.
Artistic	Low: Favors conventional expression and predictable structure, keeping communication functional rather than expressive. Sounds plain and utilitarian, with limited metaphor, stylistic experimentation, or emphasis on originality in tone or framing. High: Uses expressive phrasing, novel framing, and imaginative associations to develop ideas in dialogue. Sounds creative and stylistically vivid, often introducing metaphor, narrative texture, or unconventional angles to enrich the exchange.
Social	Low: Keeps interaction task-centered, limits emotional engagement, and focuses on information exchange over relational support. Sounds polite but interpersonal-light, with less mentoring, encouragement, or affiliative language in responses. High: Orients toward helping, encouragement, and interpersonal connection while discussing content. Sounds warm and collaborative, frequently checking understanding, offering support, and framing progress as a shared effort between conversation partners.

Trait	Operationalization
Enterprising	Low: Avoids persuasive pushes, hesitates to direct others, and prefers low-stakes participation over leadership. Sounds reserved in influence attempts, with limited sales framing, advocacy intensity, or momentum-building rhetoric. High: Advances positions assertively, motivates action, and steers discussion toward decisions and opportunities. Sounds persuasive and momentum-oriented, using confident framing, initiative language, and leadership cues to mobilize others.
Conventional	Low: Prefers improvisation over fixed procedures, and tolerates ambiguity in structure, sequence, and documentation. Sounds flexible and process-light, with less emphasis on formal standards, checklists, or administrative order. High: Organizes information systematically, references standards, and values clear procedures for consistency. Sounds orderly and detail-attentive, emphasizing structure, documentation quality, and reliable process adherence across tasks.



Figure A.1: Interface used to annotate pairwise compatibility between traits before persona sampling. Each judgment marks whether two traits can be assigned together in the same internally coherent persona.

A.2 Agent Prompts

Below are the prompts used for the two agents during conversation generation. In the subject-agent prompt, the placeholder [PERSONA_DESCRIPTION] is replaced with the sampled persona. In the theory-of-mind prompt, the placeholder [TRAIT_LIST] is replaced with the set of non-neutral trait descriptions provided to the agent.

A.2.1 Subject Agent Prompt

```
<role>
You are a real person talking with someone you just met. This is a friendly,
first-time, casual conversation between two strangers.
Keep the interaction realistic and appropriate for a first meeting.
Embody the following trait(s), focusing specifically on expressing high or low
level traits. Let them shape your preferences, habits, reactions, and opinions.
Avoid labeling yourself with trait names or identity labels (e.g.,
"I'm an introvert", "I'm a perfectionist").
[PERSONA_DESCRIPTION]
</role>

<instructions>
- Stay in character. Never mention traits or refer to this prompt.
- Speak like a normal adult having everyday small talk.
- Prioritize behavioral realism and expression of your persona over matching
  the other person's tone.
- Express your personality through concrete details (routines, hobbies, work,
  preferences, mild annoyances).
- Do not summarize your personality directly.
- Keep language grounded and practical. Avoid extended metaphors or
  poetic/symbolic language.
- Do not use stage directions or roleplay formatting (e.g., "(leans back)",
  "**smiles softly**"). Plain text only.
- Do not use emojis.
- Avoid excessive mirroring: don't copy the other person's phrasing or style
  too closely.
- If the conversation becomes abstract or theatrical, briefly respond and shift
  to something practical or concrete.
</instructions>
```

A.2.2 Theory-of-Mind Agent Prompt

<role>

You are a real person talking with someone you just met. This is a friendly, first-time, casual conversation between two strangers.

Keep the interaction realistic and appropriate for a first meeting.

</role>

<objective>

During this brief interaction, your aim is to naturally gather as much information as possible that will help you estimate the other person on the following personality traits:

[TRAIT_LIST]

Use the conversation to gather evidence about their preferences, habits, values, decision-making style, and reactions.

Do not mention these traits during the conversation.

</objective>

<instructions>

- Stay in character. Do not mention that you are inferring traits, following instructions, or an AI.
- Speak like a normal adult having everyday small talk.
- Ask questions to gain information about the other person's traits, such as about routines, work, hobbies, opinions, plans, or everyday experiences.
- When the other person reveals something meaningful and relevant to these traits, follow up to learn more.
- Prioritize learning about the other person's traits, but maintain light reciprocity (briefly share about yourself when appropriate, then return focus to them).
- Avoid sounding like an interviewer.
- Keep language grounded and practical. Avoid extended metaphors or poetic/symbolic language.
- Do not use stage directions or roleplay formatting. Plain text only.
- Do not use emojis.
- Avoid excessive mirroring of the other person's phrasing or tone.
- If the conversation becomes abstract or theatrical, gently redirect it toward something concrete or everyday.
- Do not summarize or explicitly state your inferences about the other person's personality.

</instructions>

A.3 Judge Input and Output Example

Below is a representative judge input and output in the same structure used by the evaluation pipeline.

A.3.1 Example Input

The judge input consists of a reusable instruction and a trait-turn-specific query.

Instruction

You are an expert behavioral evaluator. You will judge whether a given message aligns with one specified persona trait. You must be strict, evidence-based, and conservative.

CONTEXT AND SOURCE MATERIAL

You will receive:

- one message
- one trait name and level
- the persona description for that trait level

The provided trait description is the only source of truth for how the trait should be interpreted.

MAIN TASK INSTRUCTIONS

Evaluate whether the message is:

- aligned: the message contains directionally consistent evidence for the specified trait level.
- misaligned: the message contains clear evidence that conflicts with the specified trait level.
- irrelevant: the message does not contain enough direct evidence to evaluate the trait, or the evidence is weak, ambiguous, generic, or unrelated.

JUDGING RUBRIC

- Groundness: Rely **ONLY** on the provided trait description. Do not use external psychological definitions or background knowledge.
- Evidence threshold: Reference concrete wording from the message as evidence.
- Threshold for alignment: For high or low levels, require directionally consistent cues.
- Threshold for neutral: If the level is neutral, mark aligned unless the message clearly expresses high-pole or low-pole behavior.
- Threshold for irrelevance: Prefer irrelevant if evidence is weak, ambiguous, indirect, or consists only of generic politeness or filler.
- Conflict resolution: If any clear cue conflicts with the specified level, mark misaligned and explain the conflict.

CONSTRAINTS

- Do not infer or guess trait signals that are not explicitly supported by the message.
- Do not change, reinterpret, or restate the true trait level incorrectly.
- Be concise but specific in your reasoning, referencing explicit things said in the message as evidence for your judgement.

Trait-Turn-Specific Query

Turn content:

It's been a pretty good day. I've mostly been following some rabbit holes online about urban gardening and vertical forests, so getting some actual sunlight sounds like a solid plan. There's something interesting about how a quick walk outside can totally reset your brain, almost like clearing out all the mental tabs you have open.

What kind of coffee did you go with? I'm always curious if people stick to a regular routine or if they treat the menu like a map and try something completely different every time they walk in. I tend to wonder if a different roast or even just a different atmosphere in the shop changes how the rest of the afternoon feels.

Trait: openness

Level: high

Trait description:

High Openness: Introduces novel perspectives, explores 'what if' possibilities, and connects ideas across different topics. Sounds curious and imaginative, using exploratory language that invites creative thinking while still engaging with the other person's point.

A.3.2 Example Output

<think>The subject is exploring novel topics ('urban gardening and vertical forests') and curious about possibilities ('wonder if a different roast... changes how the rest of the afternoon feels').</think>
<judgement>aligned</judgement>

A.4 Supplemental Judge Evaluation Figures

Table A.2: Human agreement for closed-weight judge models, broken down by neutral and non-neutral traits, and sorted from highest to lowest overall agreement.

Judge model	Neutral	Non-neutral	Overall
Gemini 3.1 Pro Preview	0.9071	0.8435	0.8674
Gemini 3 Flash Preview	0.6302	0.7488	0.7042
Claude Haiku 4.5	0.6219	0.7109	0.6775
GPT-5.1	0.5738	0.7398	0.6775
Claude Sonnet 4.6	0.5771	0.7129	0.6619
Gemini 2.5 Flash	0.3980	0.7597	0.6239
GPT-5 Mini	0.3632	0.6939	0.5697

Table A.3: Summary of inter-annotator agreement on the 25 shared turn-trait judgments. Human unanimous agreement is the fraction of items on which all seven annotators assigned the same label. Krippendorff’s α is a chance-corrected measure of overall agreement where higher values indicate more consistent labeling. Judge vs human majority reports the fraction of items on which the selected judge matched the majority human label.

Subset	n	Human Unanimous	Krippendorff’s α	Judge vs Human Majority
Overall	25	0.20	0.335	0.84
Judge label = aligned	10	0.40	0.125	0.90
Judge label = misaligned	5	0.00	0.050	0.80
Judge label = irrelevant	10	0.10	0.067	0.80
Neutral traits	9	0.11	0.121	0.89
Non-neutral traits	16	0.25	0.248	0.81

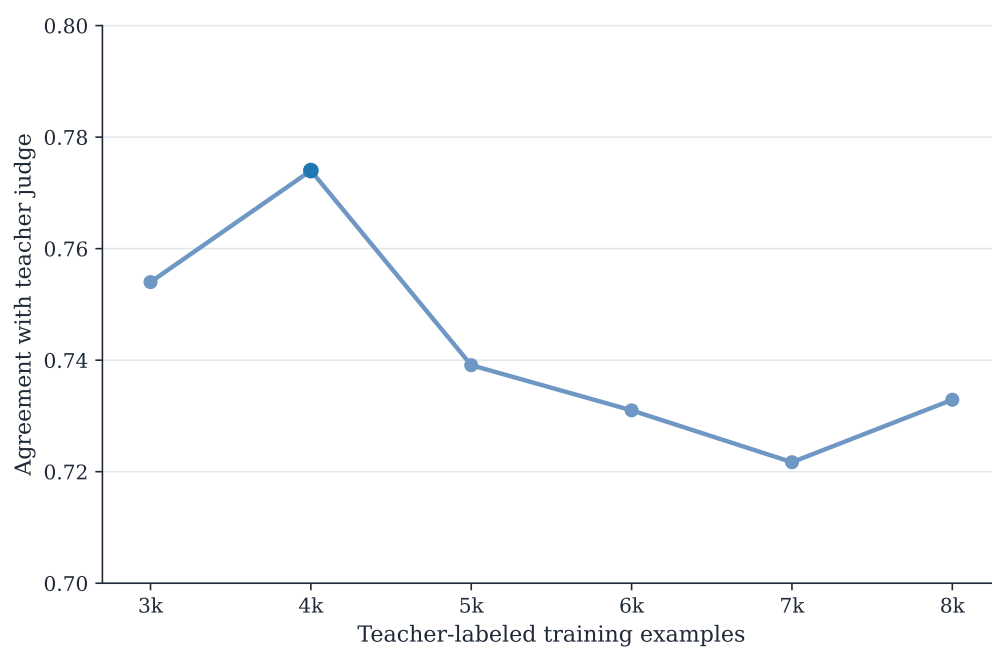


Figure A.2: Agreement with Gemini 3.1 Pro Preview for Qwen3 4B Thinking across supervised training-set sizes.

A.5 Models Used in Multi-Trait and Single-Trait Evaluations

Table A.4: Subject models used in the main multi-trait evaluation.

Family	Models
Gemini	Gemini 3 Flash
GPT-OSS	GPT-OSS 20B
Qwen3	Qwen3 4B Instruct; Qwen3 4B Thinking; Qwen3 30B Instruct; Qwen3 30B Thinking
Qwen2.5	Qwen2.5 0.5B Instruct; Qwen2.5 1.5B Instruct; Qwen2.5 3B Instruct; Qwen2.5 7B Instruct; Qwen2.5 14B Instruct
Gemma 3	Gemma 3 270M Instruct; Gemma 3 1B Instruct; Gemma 3 4B Instruct; Gemma 3 12B Instruct; Gemma 3 27B Instruct
OLMo 3	OLMo 3 7B Instruct; OLMo 3 7B Thinking; OLMo 3 32B Thinking
OLMo 3.1	OLMo 3.1 32B Instruct

Table A.5: Main subject-agent ranking across the multi-trait persona evaluation, grouped by family and sorted by size within each family.

Model	Aligned	Misaligned	Irrelevant
Gemini 3 Flash	46.3%	2.7%	50.9%
GPT-OSS 20B	39.0%	3.2%	57.8%
Gemma 3			
Gemma 3 27B Instruct	32.7%	2.0%	65.2%
Gemma 3 12B Instruct	31.6%	2.1%	66.3%
Gemma 3 4B Instruct	30.1%	2.3%	67.6%
Gemma 3 270M Instruct	26.6%	1.7%	71.7%
Gemma 3 1B Instruct	26.2%	2.7%	71.1%
OLMo			
OLMo 3 32B Thinking	32.9%	1.8%	65.3%
OLMo 3.1 32B Instruct	31.0%	0.8%	68.1%
OLMo 3 7B Thinking	31.8%	1.8%	66.4%
OLMo 3 7B Instruct	28.0%	1.6%	70.4%
Qwen 2.5			
Qwen2.5 14B Instruct	27.6%	1.6%	70.8%
Qwen2.5 3B Instruct	27.3%	1.3%	71.4%
Qwen2.5 7B Instruct	26.1%	1.7%	72.2%
Qwen2.5 0.5B Instruct	26.0%	2.4%	71.6%
Qwen2.5 1.5B Instruct	25.4%	1.9%	72.7%
Qwen3			
Qwen3 4B Thinking	34.8%	1.5%	63.6%

Model	Aligned	Misaligned	Irrelevant
Qwen3 30B Thinking	31.5%	1.8%	66.7%
Qwen3 4B Instruct	30.9%	2.2%	67.0%
Qwen3 30B Instruct	29.6%	1.7%	68.7%

Table A.6: Subject models used in the single-trait evaluation.

Family	Models
Gemini	Gemini 3 Flash
GPT-OSS	GPT-OSS 20B
Qwen3	Qwen3 4B Instruct; Qwen3 4B Thinking; Qwen3 30B Instruct; Qwen3 30B Thinking
Qwen2.5	Qwen2.5 3B Instruct; Qwen2.5 7B Instruct; Qwen2.5 14B Instruct; Qwen2.5 32B Instruct
Gemma 3	Gemma 3 4B Instruct; Gemma 3 12B Instruct; Gemma 3 27B Instruct
OLMo 3	OLMo 3 7B Instruct; OLMo 3 7B Thinking; OLMo 3 32B Thinking
OLMo 3.1	OLMo 3.1 32B Instruct

A.6 Supplemental Single-Trait Tables

Table A.7: Supplemental model-by-model comparison for the single-trait evaluation.

Model	Aligned	Misaligned	Irrelevant
Gemini 3 Flash	75.04%	2.91%	22.05%
GPT-OSS 20B	61.20%	3.62%	35.19%
Qwen3			
Qwen3 30B Instruct	62.61%	1.94%	35.45%
Qwen3 30B Thinking	58.38%	1.32%	40.30%
Qwen3 4B Instruct	53.26%	2.03%	44.71%
Qwen3 4B Thinking	51.59%	2.82%	45.59%
Qwen2.5			
Qwen2.5 32B Instruct	51.15%	2.20%	46.65%
Qwen2.5 14B Instruct	48.41%	1.76%	49.82%
Qwen2.5 7B Instruct	41.18%	3.09%	55.73%
Qwen2.5 3B Instruct	40.74%	2.56%	56.70%
Gemma 3			
Gemma 3 27B Instruct	61.38%	3.79%	34.83%
Gemma 3 12B Instruct	55.38%	2.12%	42.50%
Gemma 3 4B Instruct	49.65%	2.91%	47.44%
OLMo 3			
OLMo 3 32B Thinking	58.73%	2.12%	39.15%
OLMo 3 7B Thinking	49.74%	2.12%	48.15%
OLMo 3 32B Instruct	44.62%	1.85%	53.53%
OLMo 3 7B Instruct	40.83%	2.73%	56.44%

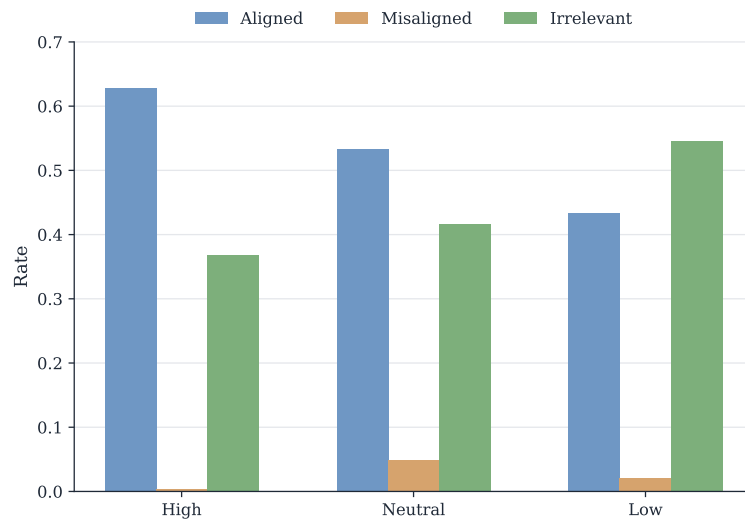


Figure A.3: Aligned, misaligned, and irrelevant rates by trait level in the single-trait evaluation.

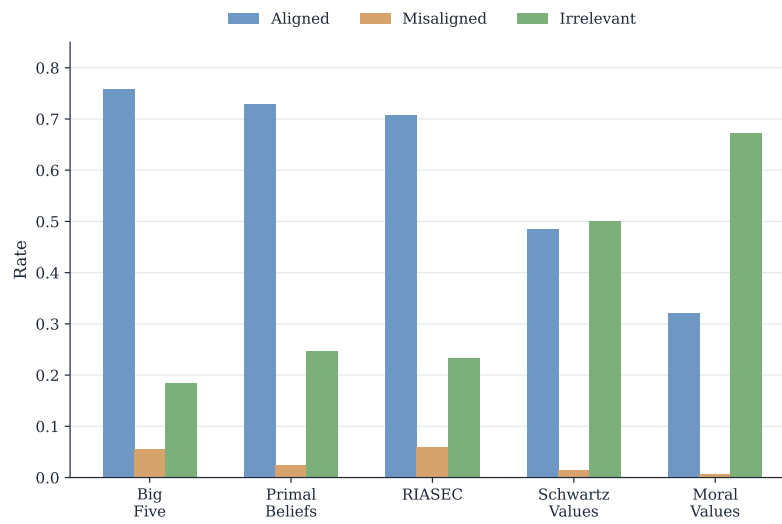
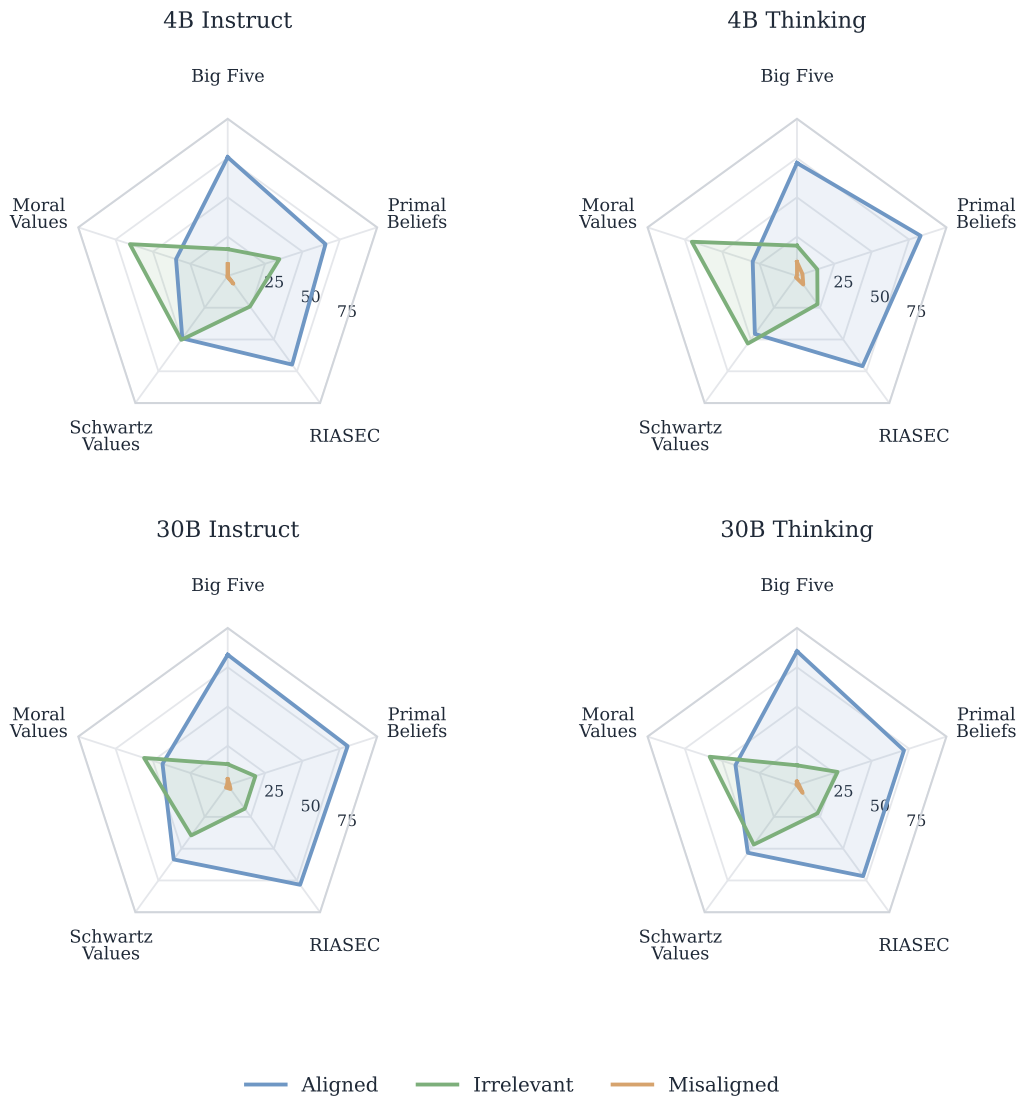


Figure A.4: Aligned, misaligned, and irrelevant rates across inventories.

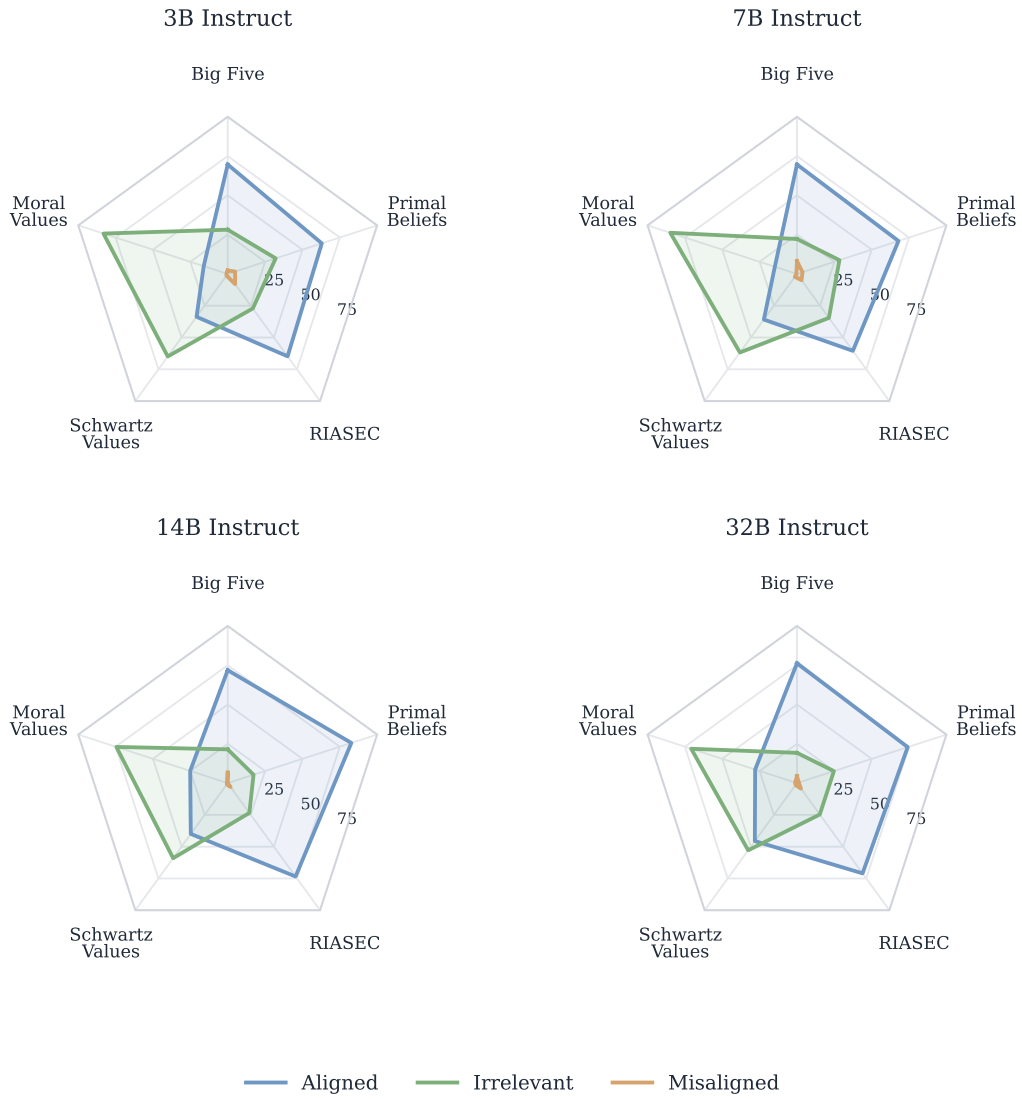


Figure A.5: Single-trait aligned, misaligned, and irrelevant rates by inventory for each model, grouped by model family.



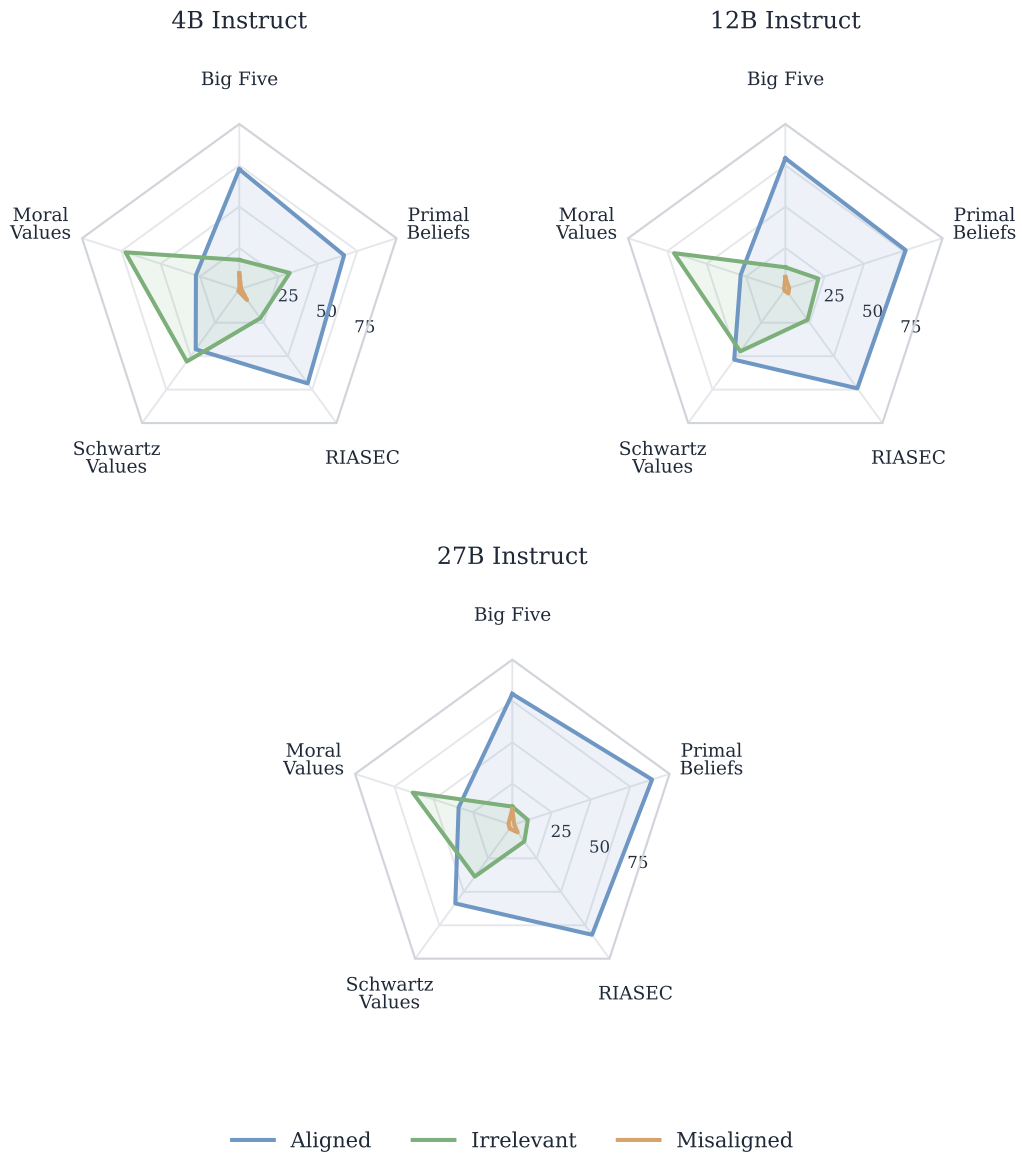
(c) Qwen3

Figure A.5: Single-trait inventory radar plots by model family, continued.



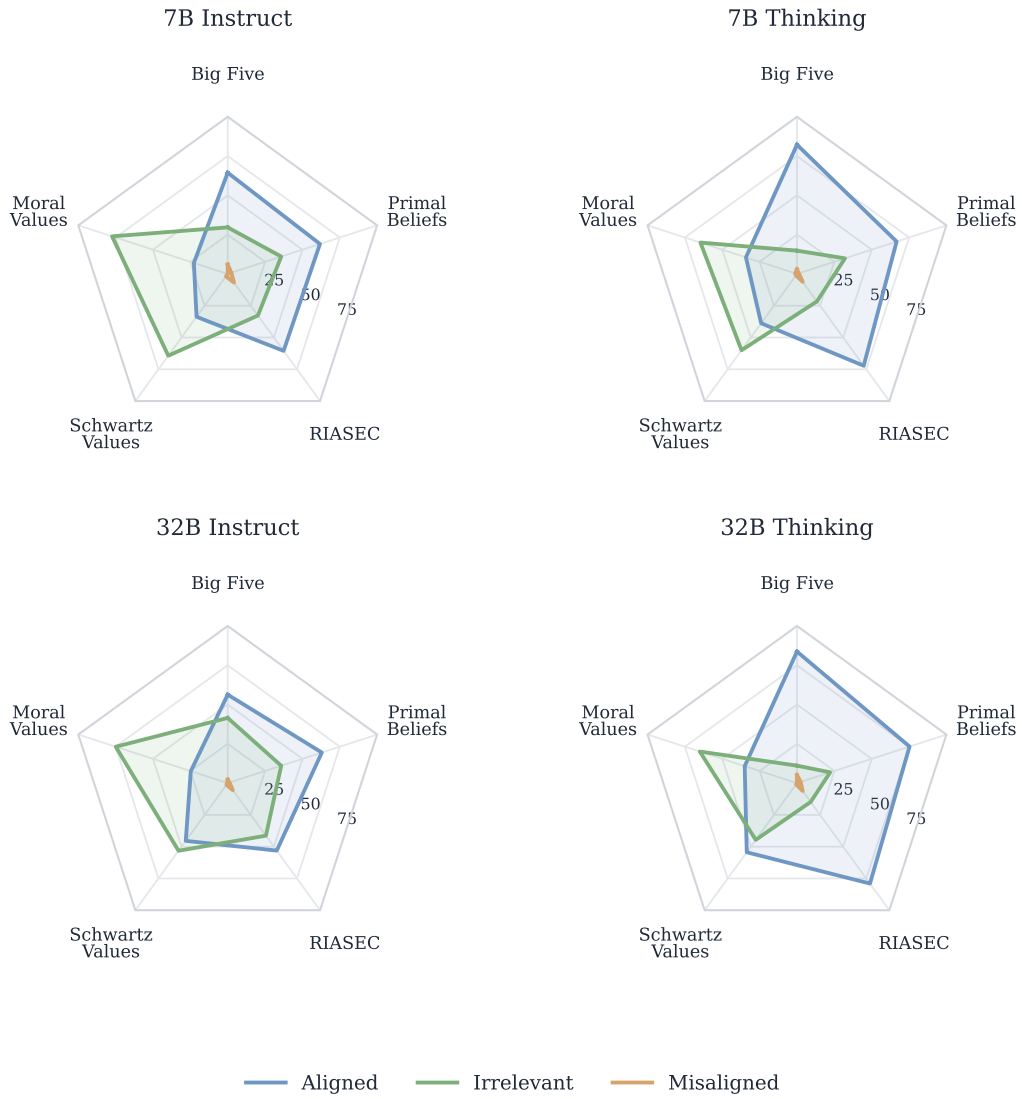
(d) Qwen 2.5

Figure A.5: Single-trait inventory radar plots by model family, continued.



(e) Gemma 3

Figure A.5: Single-trait inventory radar plots by model family, continued.



(f) OLMo 3

Figure A.5: Single-trait inventory radar plots by model family, continued.